

SECTION 7

SYSTEM DESCRIPTION

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SECTION 7

SYSTEM DESCRIPTION

7.1 GENERAL

This section contains general information that applies specifically to standard equipped helicopters serial number S-161 and subsequent. Refer to the BO 105 Maintenance Manual for information pertaining to earlier serial numbered helicopters.

7.2 LOCATION AREAS (see Fig. 7-1)

7.2.1 Nose Compartment

The nose compartment located forward of the cabin area provides a location for the battery (Model C Variants) and other electrical system components. Access is provided by a nose door fitted with an external power unit (EPU) connector access flap.

7.2.2 Cabin

The cabin area comprises the cockpit and passenger compartment.

A hinged jettisonable door on either side permits direct access to the cockpit. Large bubble-type acrylic windshields and clear acrylic windows in the crew doors and roof allow for good visibility.

A sliding door is located on either side of the fuselage for direct access to the passenger compartment. A large acrylic window is installed in each door.

7.2.3 Cargo Compartment

The cargo compartment may be separated from the cabin by a removable insulating divider. The engine deck forms the roof of the cargo compartment and acts as a firewall. Access to the compartment is from the aft end provided by two clamshell doors.

7.2.4 Avionics Equipment Bay

An avionics equipment bay is located beneath and at the aft end of the cargo compartment. This bay provides for installation of avionics equipment and is usually equipped with a forced-air ventilation system. Access is provided through a hinged and latched panel in the cargo compartment floor.

7.2.5 Tail Boom

The root of the tail boom is attached to the fuselage at an integral extension of the engine deck. The tail boom consists of a conical structure with a vertical fin. The tail rotor system

and components of both the yaw flight control and comm/nav systems are attached to the tail boom.

7.2.6 Landing Gear

The landing gear is of the fixed tubular skid-type comprised of two aluminum cross tubes passing transverse through the fuselage structure and to which aluminum longitudinal skid tubes are attached.

7.3 COCKPIT ARRANGEMENT (see Fig. 7-2)

7.3.1 Instrument Panel and Console Layout (see Fig. 7-3 thru 7-5)

The cockpit-mounted controls, indicators and annunciators shown in Figs. 7-3 thru 7-5 will be discussed later with their associated systems. Annunciator lights and their fault indications are summarized under lighting equipment. Electrical fuses referred to in this section are normally located on a fuse panel discussed later under the electrical system.

7.3.2 Crew Member Seats

Both pilot and copilot seats are adjustable fore-aft and fitted with lap safety belts.

7.4 EMERGENCY EQUIPMENT

7.4.1 Engine Overheat and Fire Detection

Two heat-sensitive detectors are located on each engine. When the ambient temperature in either engine compartment increases beyond limits, the detector contacts open and, via a control unit, energize the respective F warning light located on the instrument panel. Each system is powered from the Main Bus via a fuse.

7.4.2 Portable Fire Extinguisher

A portable fire extinguisher is normally mounted on a quick-release support on the forward edge of the pilot's seat. The fire extinguishing agent is released by a hand-operated lever on top of the extinguisher. Inadvertent discharge is prevented by a safety-pin on the actuating lever. Operating instructions are printed on the extinguisher.

7.4.3 First Aid Kit

A first aid kit is located either in a mount on the cabin floor or in a retaining net secured on the inside of the cargo compartment RH clamshell door.

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7.4.4 Emergency VHF Radio

An emergency VHF radio, P/N 105-897936 located in the cockpit beneath the pilot's collective lever, shall be operated in accordance with its appropriate instructions.

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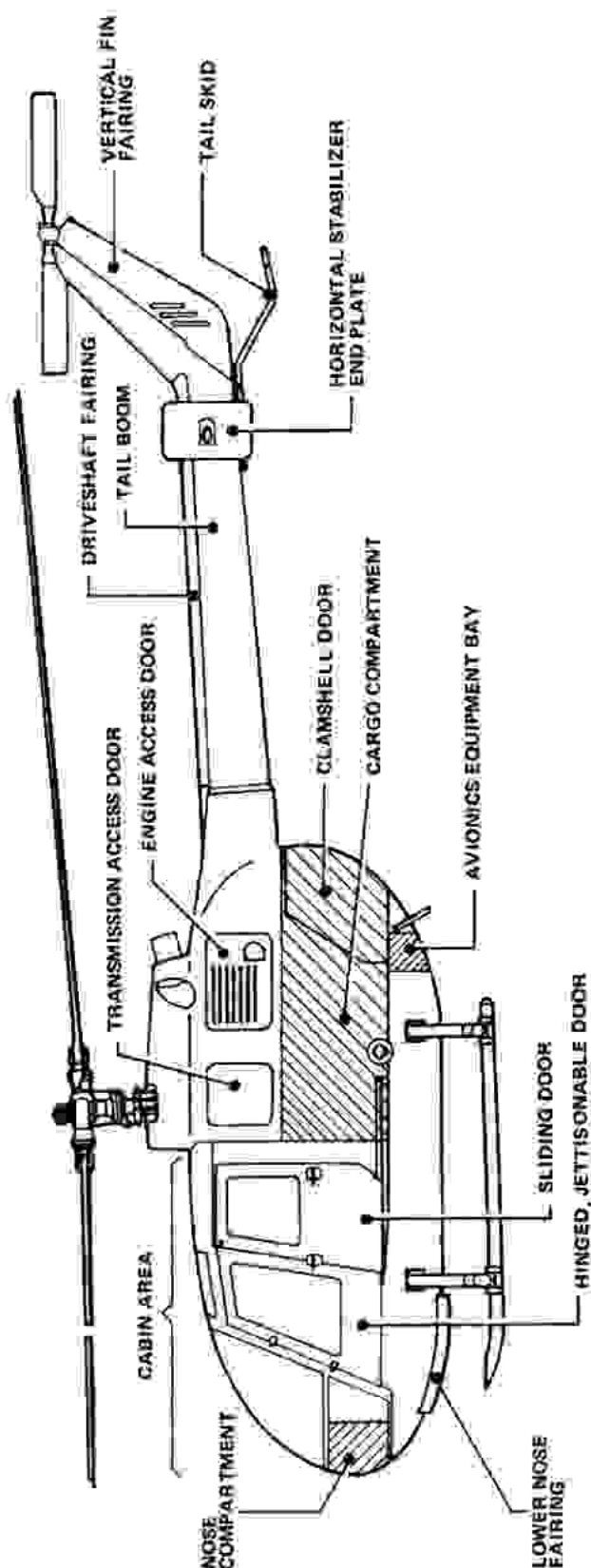
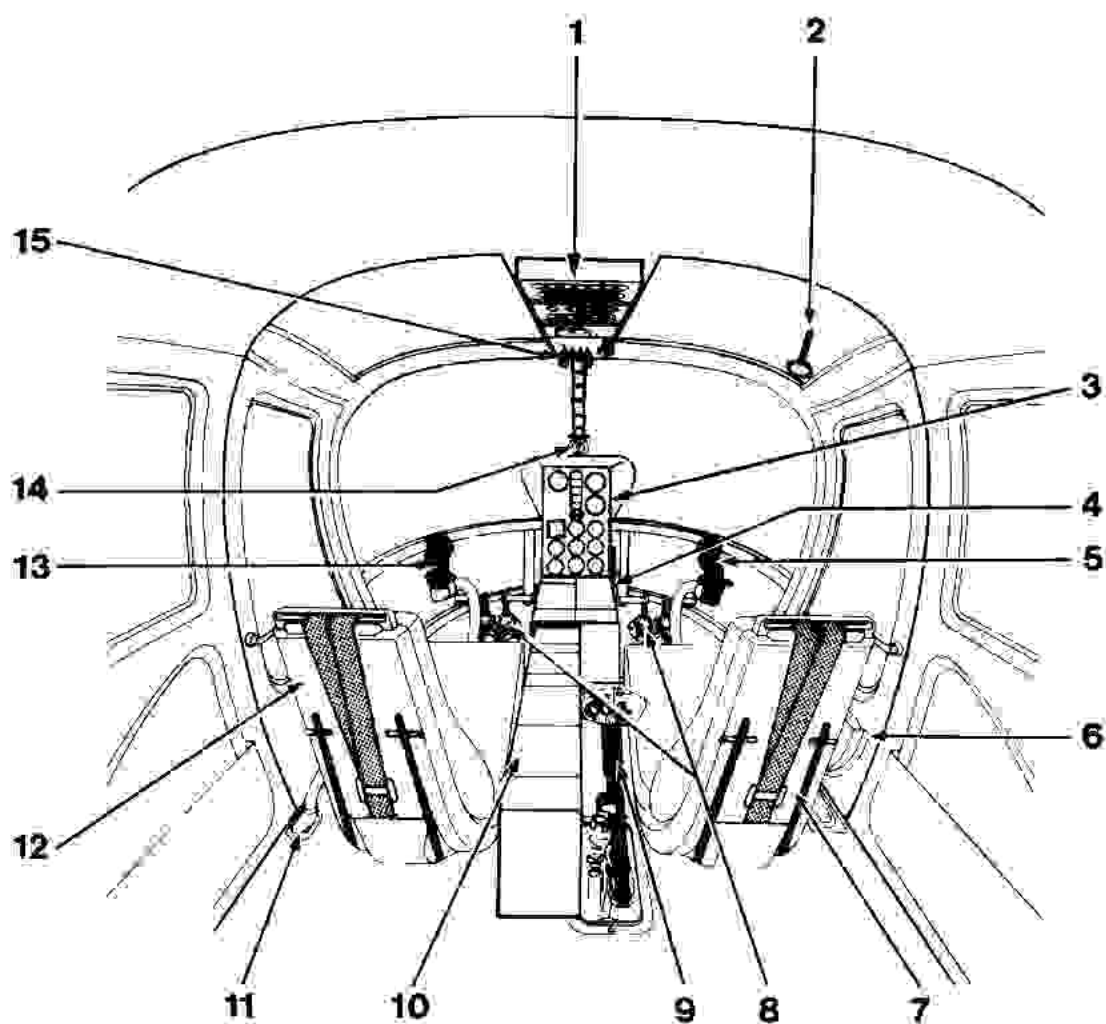


Fig. 7-1 Location Areas and Access



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|----------------------|------------------------------|-----------------------------|
| 1 OVERHEAD CONSOLE | 6 MAP POCKET | 11 COPILOT COLLECTIVE LEVER |
| 2 OAT INDICATOR | 7 PILOT SEAT | 12 COPILOT SEAT |
| 3 INSTRUMENT PANEL | 8 DIRECTIONAL CONTROL PEDALS | 13 COPILOT CYCLIC STICK |
| 4 SWITCH PANEL | 9 PILOT COLLECTIVE LEVER | 14 STANDBY MAGNETIC COMPASS |
| 5 PILOT CYCLIC STICK | 10 CENTER CONSOLE | 15 POWER LEVERS |

Fig. 7-2 Typical Cockpit Layout

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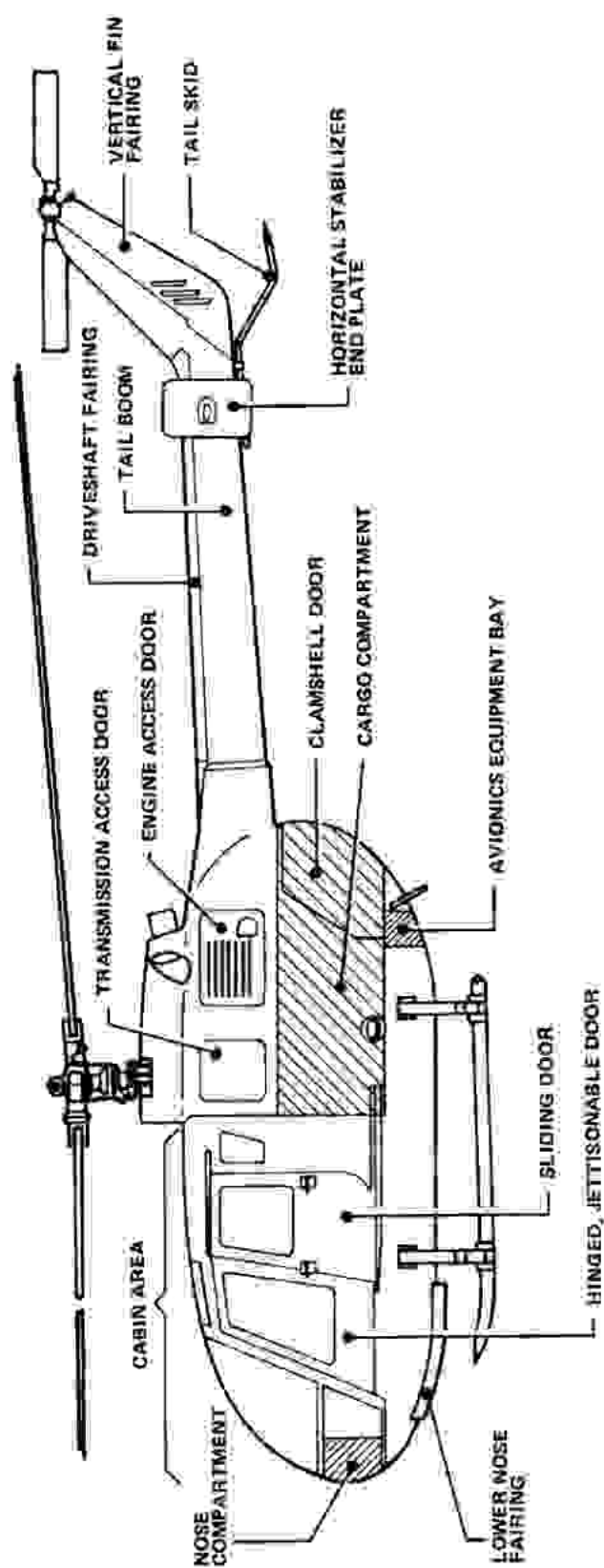
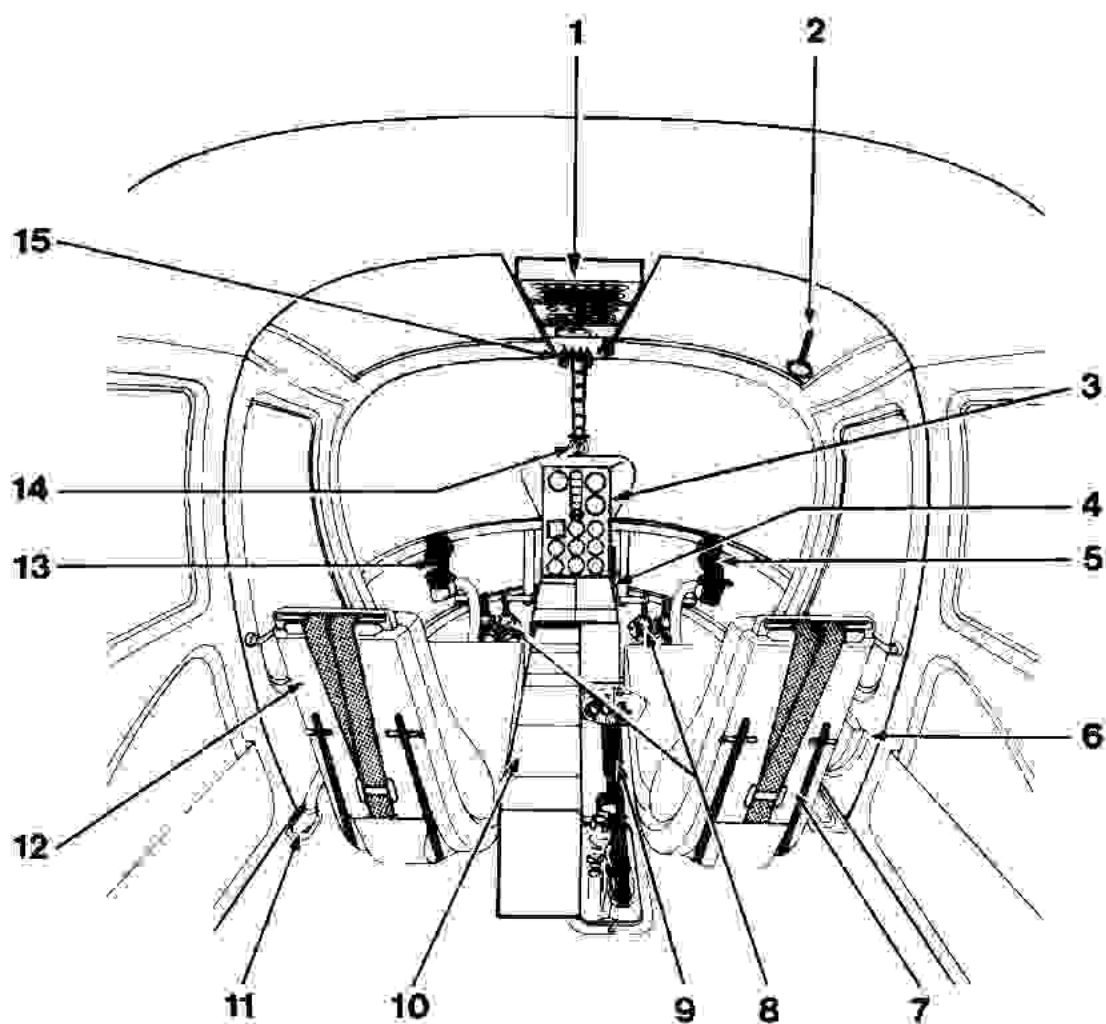


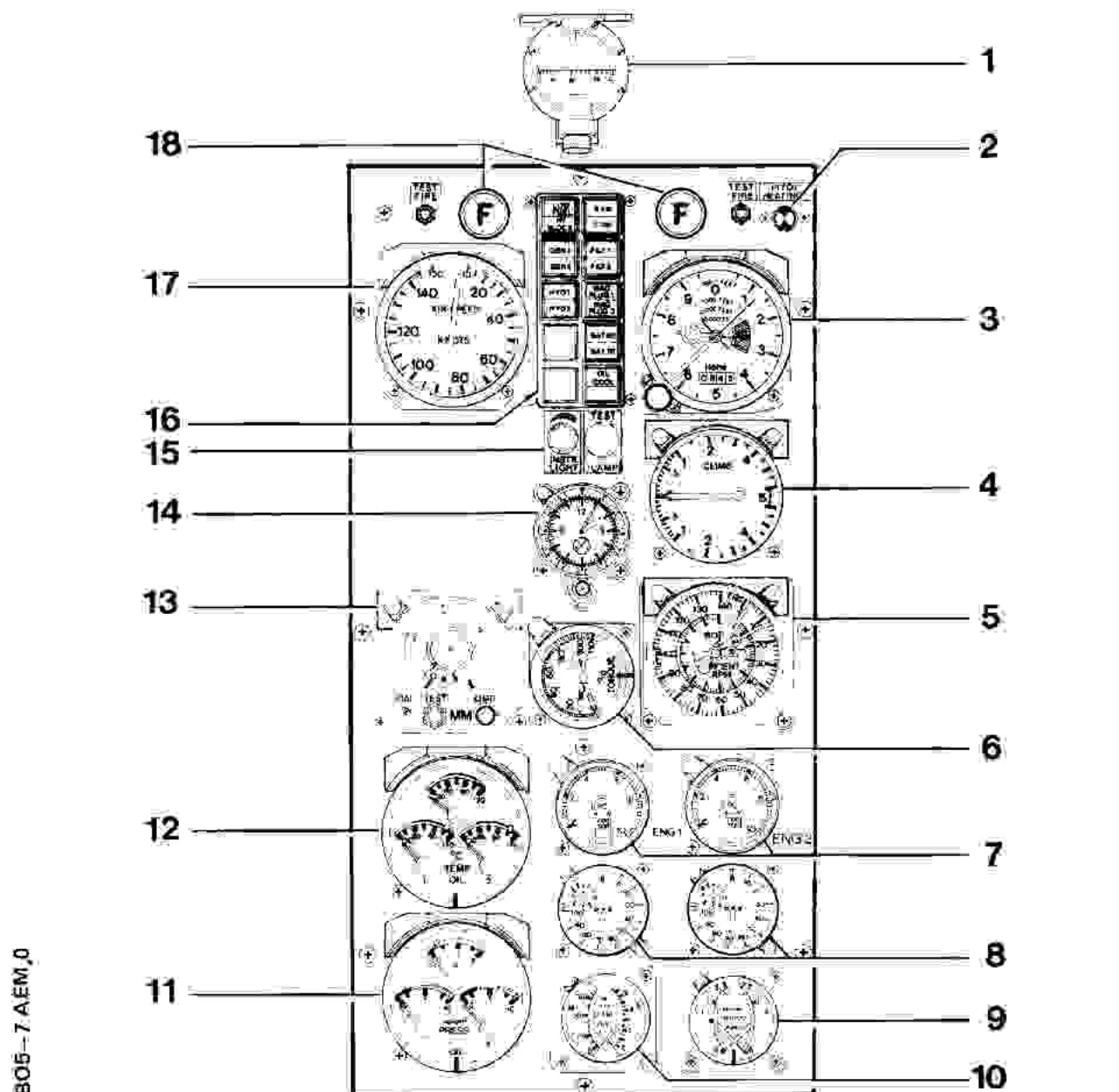
Fig. 7-1 Location Areas and Access



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- | | | |
|----------------------|------------------------------|-----------------------------|
| 1 OVERHEAD CONSOLE | 6 MAP POCKET | 11 COPILOT COLLECTIVE LEVER |
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| 5 PILOT CYCLIC STICK | 10 CENTER CONSOLE | 15 POWER LEVERS |

Fig. 7-2 Typical Cockpit Layout



- | | | |
|-----------------------------|--------------------------------------|--|
| 1. STANDBY MAGNETIC COMPASS | 8. ENGINE NO. INDICATORS | 14. CLOCK |
| 2. PITOT HEAT INDICATOR | 9. FUEL PRESSURE INDICATOR | 15. INSTRUMENT LIGHT RHEOSTAT AND ANNUNCIATOR LIGHT LAMP TEST PUSHBUTTON |
| 3. ALTIMETER | 10. FUEL QUANTITY INDICATOR | 16. WARNING LIGHT SEGMENTS |
| 4. VERTICAL SPEED INDICATOR | 11. TRIPLE OIL PRESSURE INDICATOR | 17. AIRSPEED INDICATOR |
| 5. TRIPLE TACHOMETER | 12. TRIPLE OIL TEMPERATURE INDICATOR | 18. ENGINE FIRE WARNING LIGHTS WITH TEST FIRE PUSHBUTTONS |
| 6. DUAL TORQUE INDICATOR | 13. MAST MOMENT INDICATOR | |
| 7. ENGINE TOT INDICATORS | | |

Fig. 7-3 Typical Instrument Panel

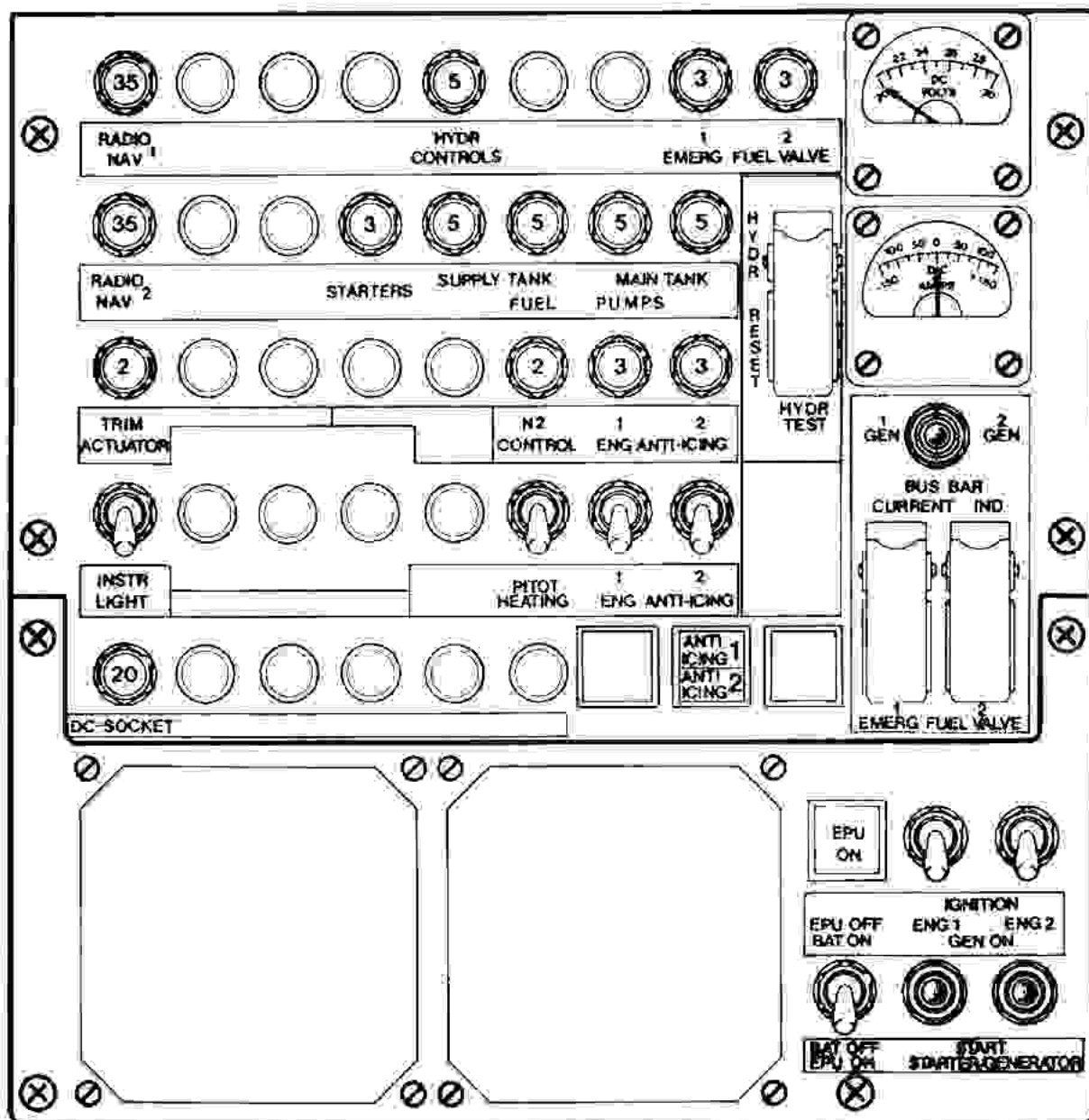
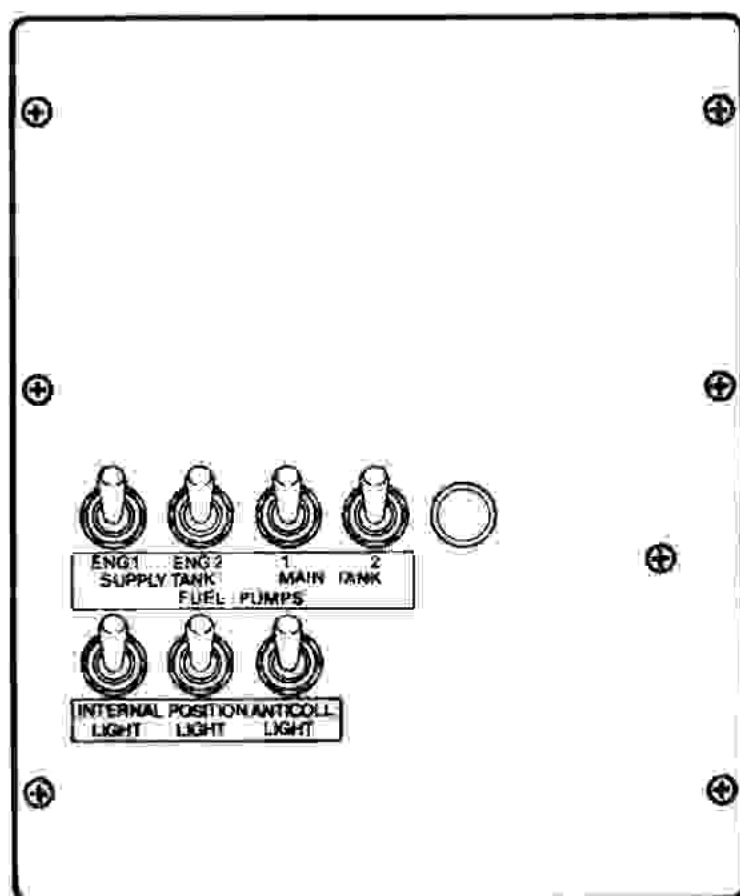


Fig. 7-4 Typical Center Console Switch Panel



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Fig. 7-5 Typical Overhead Console

7.5 POWER PLANT AND RELATED SYSTEMS

7.5.1 Engines (see Fig. 7-6)

Two Allison 250-C20 series gas turbine engines are located in separate fireproof compartments aft of the main transmission and above the cargo compartment. Each engine is divided into four modules:

- Compressor section consisting of a compressor support case assembly, a compressor and a diffuser assembly. The compressor has six axial stages and one centrifugal stage. Components mounted on the compressor section are an anti-icing air valve and a compressor bleed control valve.
- Combustion section consisting of an outer combustion case and a combustion liner. A spark igniter and a fuel nozzle are mounted in the aft end of the outer combustion case. Two external air ducts connect the compressor section with the combustion section.
- Turbine section consisting of turbine supports, collector supports, a gas producer turbine rotor and a power turbine rotor. The two stage gas producer turbine drives the compressor and N_1 accessory gear train. The two-stage power turbine furnishes the output power of the engine via the power and accessory gearbox. Mounted on the turbine section are a thermocouple harness, exhaust ducts and a fire-shield to which are attached the two fire detectors.
- Power and accessory gearbox section housing the main power and accessory gear drive trains in a single gear case. The gear case serves as the structural support of the engine. All engine driven components including the engine-mounted accessories are attached to the case. A two-stage gear set reduces the power turbine rotational speed to the output drive rotational speed. Accessories driven by the power turbine (N_2) are the N_2 tachgenerator and N_2 governor. The gas producer gear train drives the oil pump assembly (internally mounted), the fuel pump, N_1 tachgenerator, gas producer fuel control and starter/generator.

7.5.2 Air Induction

The engines receive air from the main transmission compartment which acts as a plenum chamber. An air intake fairing above the cabin roof forward of the transmission supplies a constant airflow towards the transmission compartment.

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A deflector shield is installed within the air intake fairing to direct the airflow downwards thus preventing injection of foreign matter into the engines.

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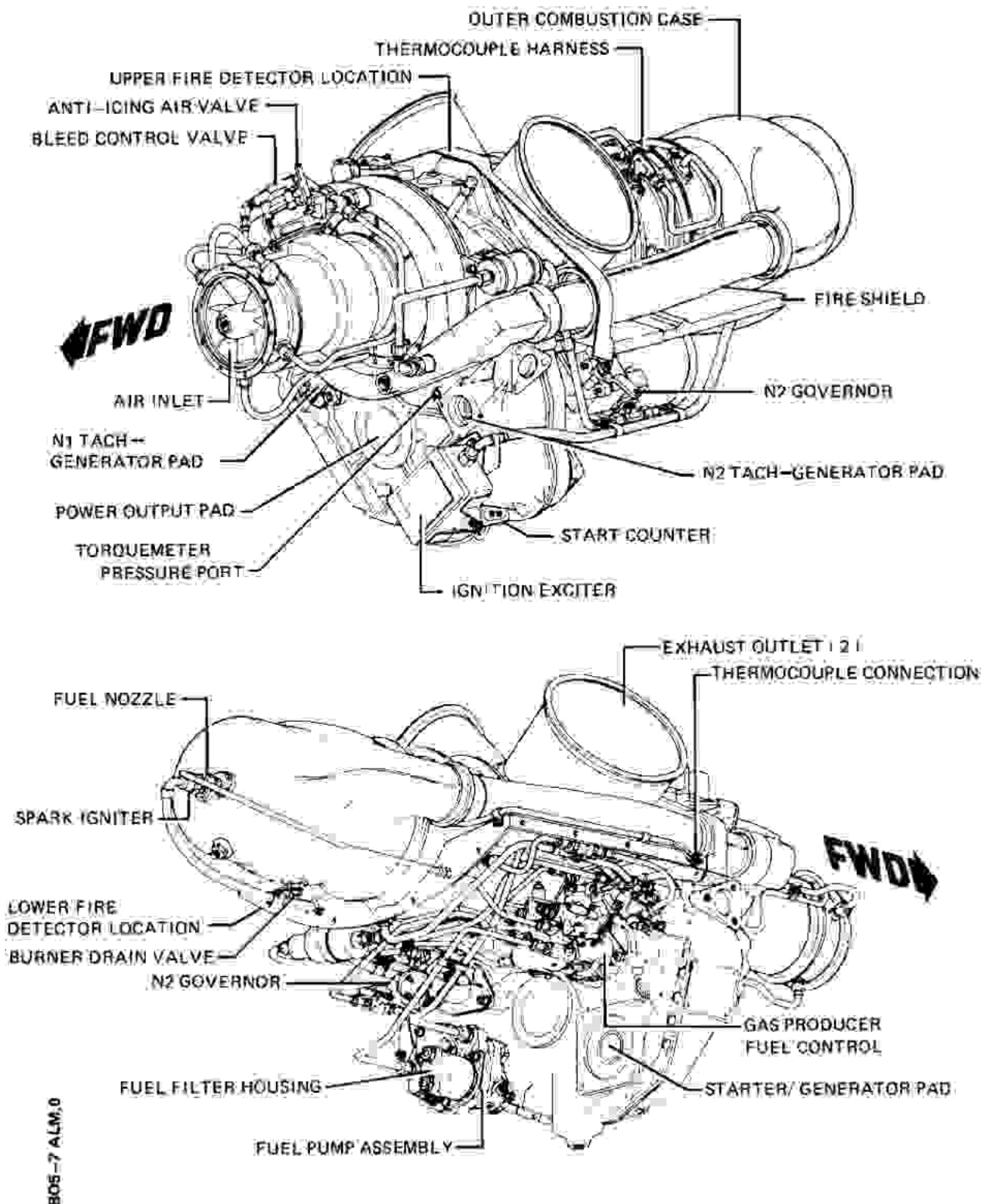


Fig. 7-6 Engine Component Locations

7.5.3 Inlet Anti-icing System

To accomplish engine anti-icing functions, hot compressor discharge air is tapped from the diffuser scroll through an electrical actuator operated anti-icing valve and routed via external lines to the compressor inlet guide vanes and front bearing support hub. The electrical actuators are controlled by the ENG ANTI-ICING - 1, 2 switches and monitored by the **ANTI ICING 1 (2)** caution lights all located on the center console switch panel.

In the ON position, the ENG ANTI-ICING switches supply power from the Main Bus via ENG ANTI-ICING - 1, 2 circuit breakers to the respective actuator motor retracting the actuator arm and opening the anti-icing valve.

A limit switch stops the actuator motor and activates the respective **ANTI ICING** caution light. The caution light remains on until after the system is switched OFF and the actuator has fully closed the anti-icing valve.

7.5.4 Fuel and Control System

Each engine fuel and control system comprises an engine-mounted fuel pump and filter assembly, a gas producer fuel control, a power turbine (N₂) governor, and a fuel nozzle. This system controls engine power output by controlling the gas producer speed. Gas producer speed levels are established by the action of the power turbine fuel governor which senses power turbine (N₂) speed. The power turbine (load) speed is selected by the pilot (ENGINE TRIM switch) and the power required to maintain this speed is automatically maintained by power turbine governor action on metered fuel flow.

A differential pressure switch is connected to the engine fuel filter to detect the beginning of clogging and activate the respective **FILT 1 (2)** warning light. A clogged filter element will be bypassed automatically supplying unfiltered fuel to that engine.

7.5.5 Oil System (see Fig. 7-7)

Each engine is lubricated by an independent pressurized recirculating dry sump system. A gear-type pressure and scavenge pump assembly is mounted within the gearbox. An assembly containing an oil filter element, a filter bypass valve, and a pressure regulating valve is located in the upper RH side of the gearbox housing and is accessible from the top of the engine.

A combined dual engine oil tank and engine/transmission oil cooler assembly is located to the right of the main transmission. The forward half of the tank/cooler assembly is connected to the left engine oil system while the rear half is connected to the right engine oil system. Two engine-mounted indicating chip detectors that activate the respective **MAG PLUG 1 (2)** warning light, are installed in each system.

The operation of each oil system is completely automatic self-regulating and does not require operator control beyond monitoring the indicators.

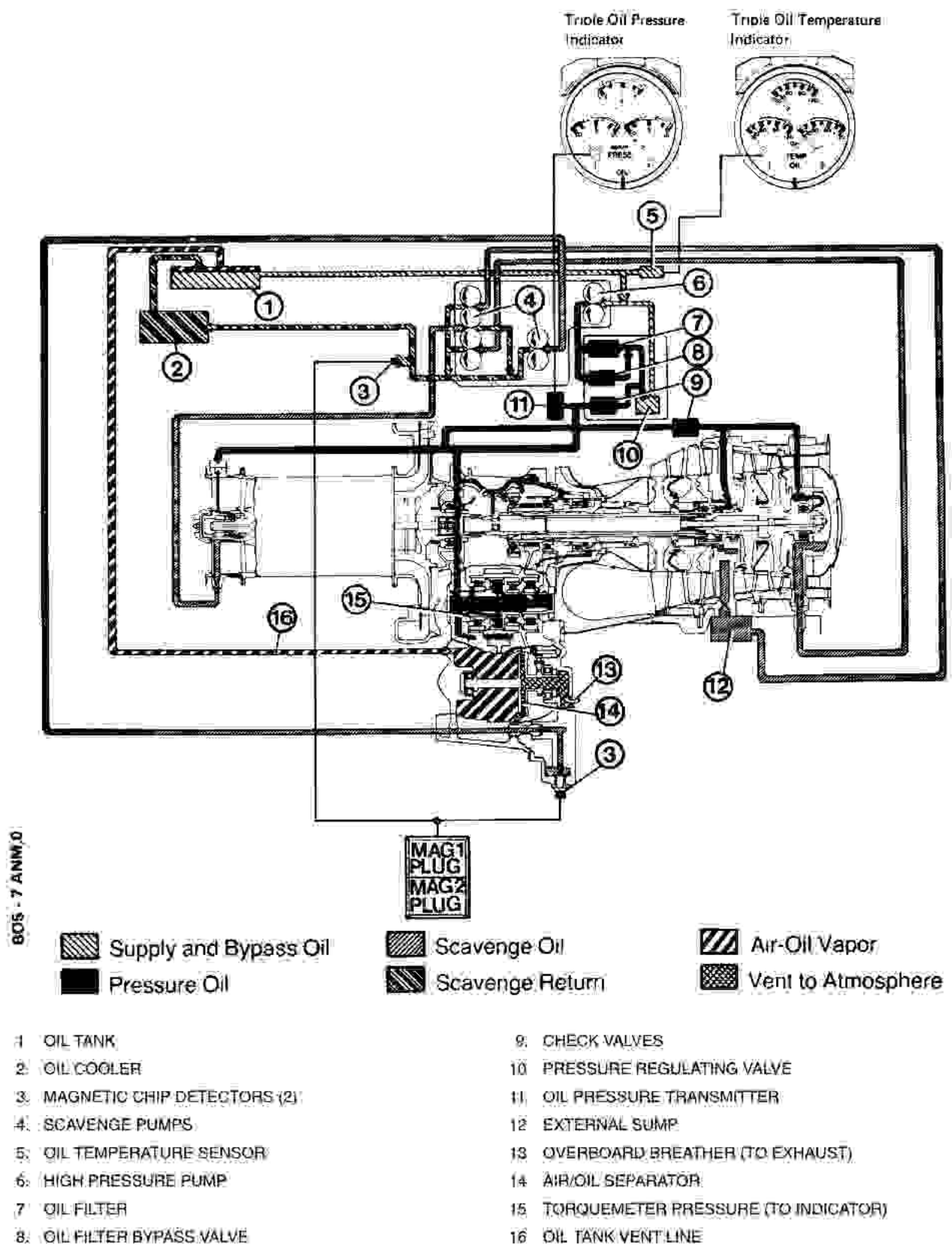


Fig. 7-7 Engine Oil System Schematic

7.5.6 Starting and Ignition System

Each engine is equipped with an independent starting/ignition system for starting the engines individually and so designed that simultaneous starting of both engines is impossible. A starter/generator unit mounted to the accessory gearbox drives the N_1 gear train when the starter motor circuits are activated. An ignition exciter unit and spark igniter comprise the ignition portion of each engine's system.

The STARTER/GENERATOR - ENG 1, ENG 2 switches on the center console switch panel are spring-loaded to the center off position when using the START position. In the START position, the switch supplies electrical power from the Main Bus via a common STARTERS circuit breaker to the respective starter relay. This relay in turn activates the starter motor circuit by supplying power from the Starter Bus and provides power to the respective IGNITION switch.

With the IGNITION switch in the on position, the exciter unit begins sending high power impulses to the spark igniter. Venting the engines is therefore possible by energizing the starter circuit without switching on the ignition circuit.

7.5.7 Control System

The engines have a conventional control system. I.e. power lever position and the degree of collective pitch basically establish the power output demands placed on the engines.

The gas producer (N_1) rpm of both engines is independently controlled by means of separate power levers mounted on the cockpit ceiling forward of the overhead console. Each power lever is mechanically connected to its respective engine gas producer fuel control to permit manual establishment of the gas producer speeds between OFF, IDLE and FLIGHT.

A separate positive lock is provided for each lever in the IDLE position so that repositioning of the levers to a position other than IDLE requires depressing the lock buttons.

Individual and simultaneous dual engine power (N_2) trimming is accomplished by means of a four-way ENGINE TRIM switch on each collective lever switch box (see Fig. 7-14). Pressing the switch to either left or right of center will increase the respective engine torque and simultaneously reduce the torque of the other engine. Pressing the switch either forward or aft will simultaneously increase or decrease respectively the torque of both engines. The pilot's ENGINE TRIM switch has command authority over the copilot's switch if installed.

A mechanical connection between the N_2 governor trim actuator and the collective control hydraulic boost actuator provides automatic rpm control (droop compensation) when changes in collective pitch occur.

7.5.8 Indicators (see Fig. 7-3.)

Torque indicator (% TORQUE) simultaneously indicates the percent of torque output of both engines. Each engine's torquemeter, incorporated in the accessory gearbox, provides an oil pressure signal directly proportional to the output torque. This oil pressure is piped directly to the respective indicator.

Triple tachometer indicator (ENG-ROTOR PERCENT RPM) simultaneously indicates both engine's power turbine speed (N_2) on an outer scale in percent rpm. The engine % N_2 rpm signals are produced by a tachgenerator mounted on the left front face of the accessory gearbox.

Turbine outlet temperature indicators ($^{\circ}\text{C} \times 100 \text{ TOT}$) indicate individual engine exhaust gas temperature measured by four thermocouple probes placed downstream of the power turbine. The bimetallic thermocouples generate DC voltage directly proportional to the sensed gas temperature.

Gas producer tachometers (% N_1) indicate individual engine gas producer speed (N_1) in percent of rated rpm. The gas producer turbine speed is measured by a tachgenerator mounted on the right front face of the accessory gearbox.

Triple oil temperature indicator ($^{\circ}\text{C TEMP OIL}$) simultaneously indicates engine 1 oil temp (left scale) and engine 2 oil temp (right scale). Each engine's oil temperature sensor is installed in an oil supply line fitting mounted on the top right front face of the accessory gearbox. The triple oil temperature indicator is powered from the Main Bus via a fuse.

Triple oil pressure indicator ($\text{kp/cm}^2 \text{ PRESS OIL}$) simultaneously indicates engine 1 oil pressure (left scale) and engine 2 oil pressure (right scale). Each engine oil pressure transmitter is fitted to a clamp securing the oil lines to the engine compartment floor. The triple oil pressure indicator is powered from the Main Bus via a fuse.

7.5.9 Annunciators

MAG PLUG 1, MAG PLUG 2 warning lights will activate when excessive metallic particles are detected by either engine-mounted chip detector.

ANTI ICING 1, ANTI ICING 2 caution lights will activate when each engine anti-icing system valve is open.

7.6 FUEL SYSTEM (see Fig. 7-8)

The fuel system comprises three subsystems, i.e. storage, supply, and indication. For refuelling and defueling procedures, refer to Section 8.

7.6.1 Storage

Fuel is stored in underfloor compartments using three bladder-type fuel cells comprising two tanks: a supply tank and a main tank. The supply tank, located immediately aft of the landing gear forward cross tube, provides fuel for the engines. Two aft cells, located forward and aft of the landing gear aft cross tube, are interconnected with a large diameter tube to form the main tank. The main tank fuel is pumped to the supply tank with excess fuel returning via two overflow ports.

The fuel cells are normally vented overboard and are drained through separate valves accessible on the fuselage underside. The main tank fuel cells are refueled through a common filler neck located on the left side of the fuselage. The filler neck is designed to accept a standard gravity-fill type nozzle.

7.6.2 Supply

Fuel from the main tank is transferred to the supply tank through electric pumps. Fuel is then further pumped from the supply tank to each engine through separate supply systems, each comprised of an electrical supply tank pump and an electrical emergency fuel shutoff valve.

The fuel pump switches, labeled SUPPLY TANK - ENG 1, ENG 2 and MAIN TANK - 1, 2 are located on the overhead console. These switches power the fuel pumps from the Main Bus via separate circuit breakers (FUEL PUMPS - SUPPLY TANK, MAIN TANK) located on the center console switch panel. The EMERG FUEL VALVE - 1, 2 switches are also located on the center console switch panel and power the fuel valves from the Main Bus via separate circuit breakers (EMERG FUEL VALVE - 1, 2). The two EMERG FUEL VALVE switches are guarded to prevent accidental usage.

7.6.3 Indicators

Dual fuel quantity indicator (FUEL QTY) simultaneously indicates both the supply tank (S) and main tank (M) fuel quantities. Two fuel quantity transmitter probes are installed, one in each tank. The indicator is powered from the Main Bus via a fuse.

Dual fuel pressure indicator (FUEL PRESS) simultaneously indicates the left engine (1) supply pressure and the right engine (2) supply pressure in kp/cm^2 . A pressure transmitter is mounted in an adapter fitting installed immediately upstream of each engine-mounted fuel filter. The indicator is powered from the Main Bus via a fuse.

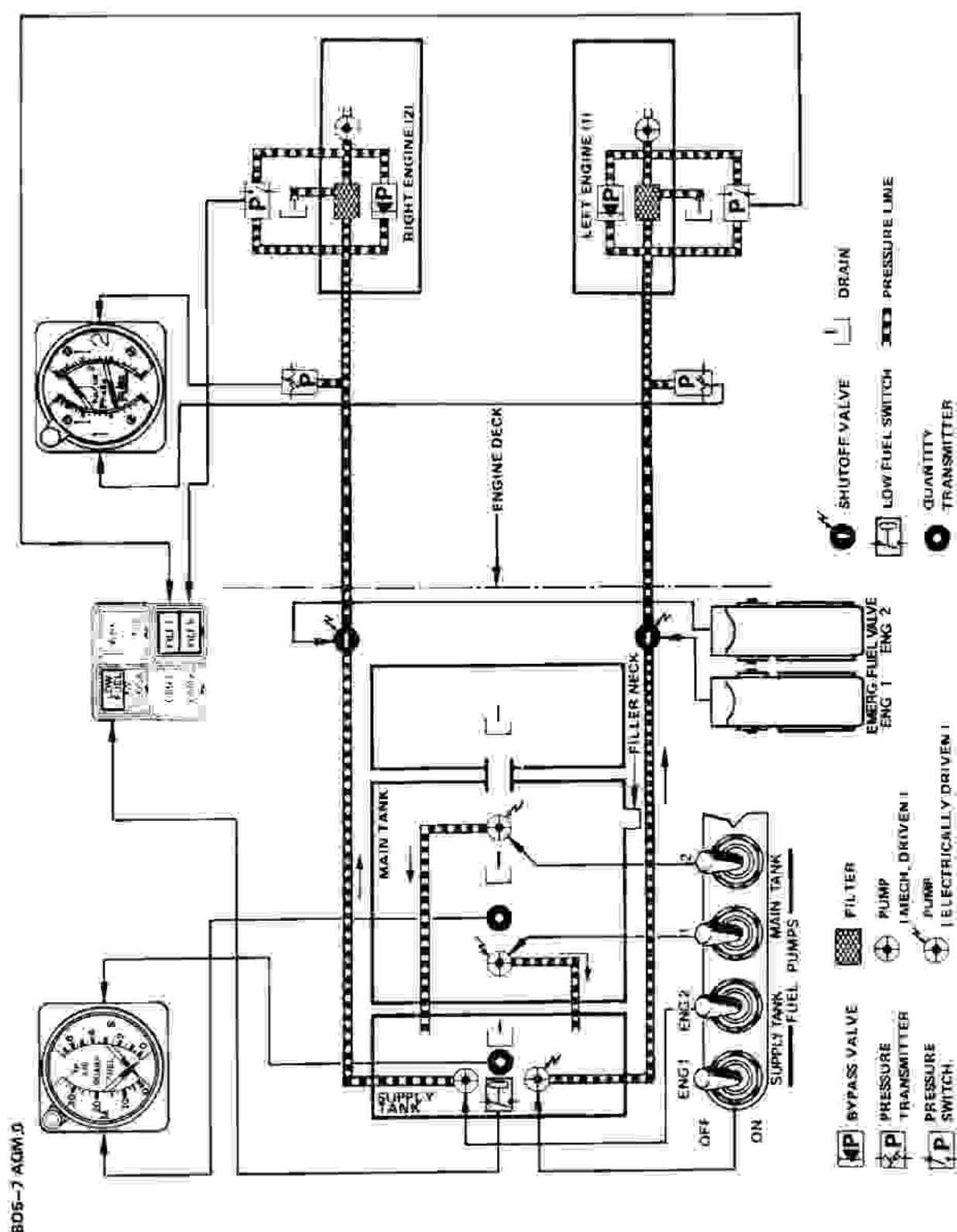


Fig. 7-8 Fuel System Schematic

7.6.4 Annunciators

Fuel system operation is monitored by the following annunciators:

- **LOW FUEL** warning light. A fuel level float switch located in the supply tank activates the **LOW FUEL** warning light when the supply tank usable quantity is below 60 kg.
- **FILT 1, FILT 2** warning lights. The **FILT 1 (2)** warning light will be activated when the engine-mounted filter becomes clogged.

7.7 POWER TRAIN (see Fig. 7-9)

The power train transmits engine power to both rotors and the transmission-mounted accessories. The power train comprises two engine drive shafts, a main transmission with accessory drives, tail rotor drive shafts, an intermediate gearbox, and a tail rotor gearbox.

7.7.1 Main Transmission (see Fig. 7-9)

Mounted forward of the engines above the cargo compartment, the main transmission combines the two engine drive shaft inputs and provides drive to the main rotor, tail rotor, accessories, and oil cooler fan. Two freewheeling units at the main transmission inputs permit either engine to be disengaged from the transmission during single-engine operation or both engines during autorotation.

Power from the engines is transmitted through various stages of reduction gearing to obtain the necessary speed for the main rotor and tail rotor. Two hydraulic pumps and a rotor rpm (N_{R0}) tachgenerator are driven by the accessory gearbox mounted on the transmission left side. Mounted on the transmission right side is an oil cooler blower fan and reduction gearbox assembly.

7.7.2 Main Transmission Oil System

The main transmission oil system lubricates and cools transmission components. An oil pump mounted on the bottom right of the transmission, suctions oil from the sump through a filter screen and feeds it then to a fine-mesh filter mounted on the forward side of the oil cooler assembly. The filter incorporates a bypass to ensure oil circulation in the event of filter clogging. Oil from the filter then flows to the oil cooler which is fitted between the two engine oil coolers. The cooled oil is then returned to a transmission case manifold fitting located on the aft side below the tail rotor drive and between the two input drives. An external line from the manifold fitting supplies oil for the pressure sensors. Passages within the transmission casings distribute oil to jets used in spraying critical components.

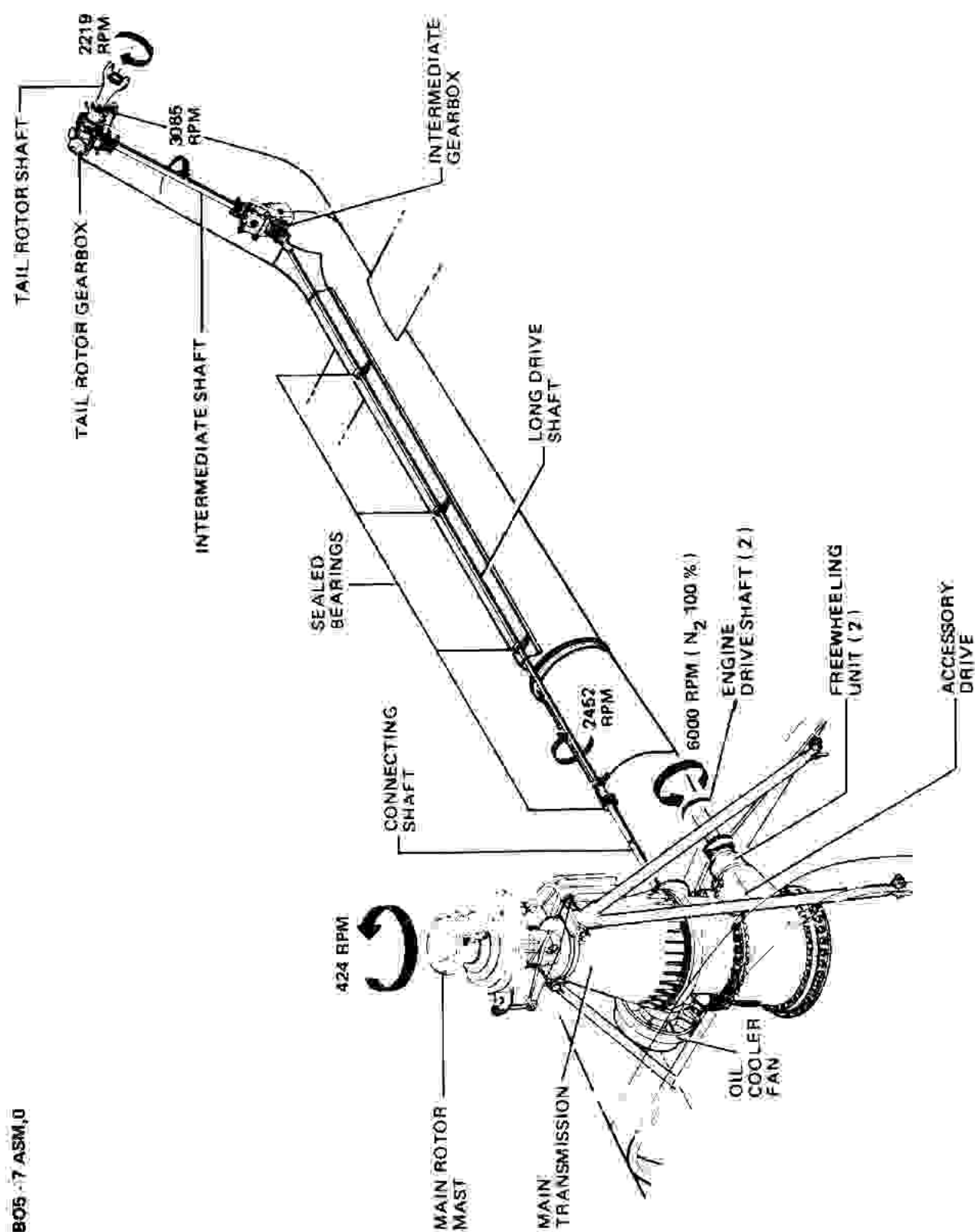


Fig. 7-9 Power Train

7.7.3 Main Transmission Indicators (see Fig. 7-10)

Triple tachometer indicator (ENG-ROTOR PERCENT RPM) indicates rotor speed (N_{R0}) on an inner scale in percent rpm. The rotor rpm signals are produced by a tachgenerator mounted on the aft face of the transmission accessory drive.

Triple oil temperature indicator ($^{\circ}\text{C}$ TEMP OIL) upper scale labeled R, indicates the transmission oil temperature. The temperature sensor is mounted on a transmission oil pump outlet manifold.

Triple oil pressure indicator (kp/cm^2 PRESS OIL) upper scale labeled R, indicates the transmission oil pressure. The oil pressure transmitter is installed in a T-fitting secured to an engine oil line hose clamp mounted on the transmission compartment floor.

7.7.4 Main Transmission Annunciators (see Fig. 7-10)

T. OIL warning light will be activated by either an overtemperature switch installed in the transmission case below the cooler fan and/or by a pressure switch installed in the T-fitting adjacent to the oil pressure transmitter. The overtemperature switch closes at an oil temperature over 105°C while the pressure switch closes whenever the oil pressure is below $0.5 \text{ kp}/\text{cm}^2$.

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OIL COOL warning light monitors operation of the oil cooler blower fan via a differential air pressure switch mounted on the oil cooler air inlet duct above the blower fan. A hose connects the switch to the air outlet shroud between the fan and oil cooler assembly. The pressure switch will activate the warning light whenever the pressure increase caused by the fan drops below a certain value.

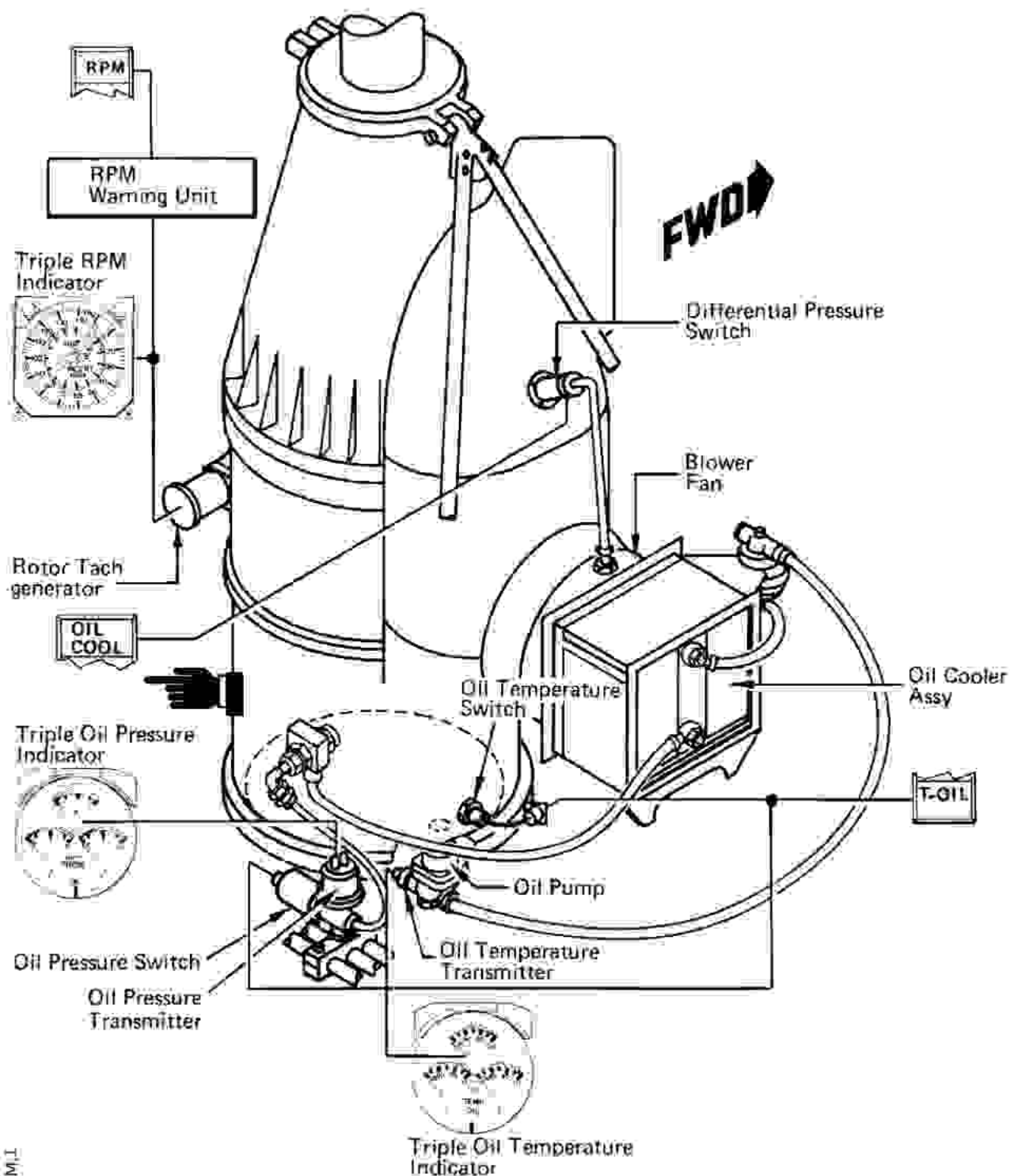
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7.7.5 Mast Moment Indication System

The mast moment indication system measures and indicates bending moments of the main rotor mast. The system is comprised of a strain gauge bridge, a rotary induction transmitter and a mast moment indicator.

Four strain gauges are bonded to the upper inner surface of the hollow rotor mast and are combined to form a bridge circuit. The rotary induction transmitter that conveys signals to and from the bridge circuit is located at the base of the main transmission.

The mast moment (MM) indicator is located on the instrument panel and is comprised of an indicator needle with colored scale, a calibration (CAL) screw, a TEST pushbutton switch and a LIMIT warning light.



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Fig. 7-10 Main Transmission Monitoring Systems

The MM indicator is powered from the Main Bus via a fuse. The indicator generates a carrier frequency which is supplied to the strain gauge bridge via the rotary transmitter. Bending moments of the rotor mast alter the resistance value of the strain gauge bridge. This modified signal is then returned through the rotary transmitter to the MM indicator where it is evaluated and then indicated by a needle deflection.

The system can be checked with the TEST pushbutton provided. When electrical power is initially applied to the system, the **LIMIT** warning light may come on. Pressing the TEST pushbutton should cause **LIMIT** warning light to illuminate and the needle to deflect to the white calibration line. Releasing the button will cancel the test indications, i.e. the **LIMIT** warning light goes off and the needle returns to the zero point (white line).

The Mast moment warning light begins to flash when the needle reaches the end of the yellow arc. It illuminates steadily when reaching the red radial. It keeps steady on at 95% of full deflection (just before reaching the white radial) even if mast moment will be reduced. It can be reset by pressing the TEST push button.

7.7.6 Tail Rotor Drive Shafts (see Fig. 7-9)

There are three tubular tail rotor drive shaft sections comprising:

- a connecting shaft with flexible couplings,
- a long shaft supported by sealed bearings and fitted with a flexible coupling at the intermediate gearbox end,
- an intermediate shaft with flexible couplings installed at the vertical fin between the intermediate and tail rotor gearboxes.

The connecting shaft is routed through a tunnel in the engine compartment dividing firewall. The long drive shaft and the intermediate shaft are covered by fairings which may be opened for maintenance and inspection.

7.7.7 Intermediate Gearbox (see Fig. 7-9)

The intermediate gearbox mounted at the base of the vertical fin increases the main transmission output rpm and changes the angle of drive 50°. The gearbox housing provides for mounting of a tail rotor yaw control bell crank. This gearbox is wet sump lubricated with an oil level sight gauge located on the left side. A magnetic-type drain plug is installed in the base of the gearbox housing.

7.7.8 Tail Rotor Gearbox (see Fig. 7-9)

The tail rotor gearbox mounted on top of the vertical fin reduces the output rpm and changes the angle of drive 90°. The tail rotor shaft passes through and is supported by the gearbox output bevel gear. The gearbox housing provides for mounting of a tail rotor yaw control bell crank. The tail rotor gearbox is also wet sump lubricated with an oil level sight gauge located on the rear face of the gearbox housing. A magnetic-type drain plug is installed at the lower rear of the housing.

7.8 ROTOR SYSTEM

7.8.1 Main Rotor (see Fig. 7-11)

The main rotor ("System Bölkow") comprises a rigid-mounted head with four glassfiber reinforced plastic (GRP) blades.

7.8.2 Main Rotor Head (see Fig. 7-11)

The rigid main rotor head is bolted directly to the main rotor mast and consists of a star hub assembly and four blade-mounting forks with pitch horns. The hub assembly consists of a single-piece cross-shaped drop forged titanium hub housing the blade retaining system with tension-torsion straps and mounting the oil reservoir and feathering hinge bearings. Each blade-mounting fork with inner sleeve assembly is inserted into an arm of the hub and retained by a tension-torsion strap. The blade-mounting forks permit blade pitch changes in response to flight control movements.

7.8.3 Main Rotor Blades (see Fig. 7-11)

Each main rotor blade is a constant-cord asymmetrical airfoil. The blades are constructed of a diagonally-layered fiberglass cloth outer skin covering a C-spar made of a unidirectional fiberglass roving lamination which forms a loop around the blade root, and a core of hard foam filler. The blade leading edge is protected against erosion and foreign object damage by titanium strips. A blade fitting assembly is bolted over the blade root and provides mounting of the blade to the head. Two trim tabs are provided on the trailing edge of the outer portion of each blade to allow for tracking adjustment.

Aerodynamic motions of lead/lag and flapping are absorbed by the blade inherent elasticity. Each blade may be fitted with pendulum absorbers near the root to reduce inflight vibrations.

7.8.4 Tail Rotor Assembly (see Fig. 7-12)

The tail rotor is of the semi-rigid teetering design. The two-bladed rotor fastened in a delta hinged hub provides antitorque action and directional control. A titanium shaft having a fork head and splined end houses two spherical teetering bearings and drives the rotating part of the yaw control system sliding sleeve. The tail rotor shaft is splined to and driven by the tail rotor output bevel gear.

Blade centrifugal forces are carried by laminated steel tension straps that attach outboard to the blade mounting forks and inboard to a strap shoe which is held centered by the tail rotor assembly mounting bolt. The blade mounting forks rotate about the hub on Teflon bushings. Dynamic weights are installed on the control yoke to reduce control pedal forces. These weights are adjusted to compensate for main rotor torque required at a hover.

The tail rotor blades are constructed similar to the main rotor blades. The blades are retained in the mounting forks by two bolts.

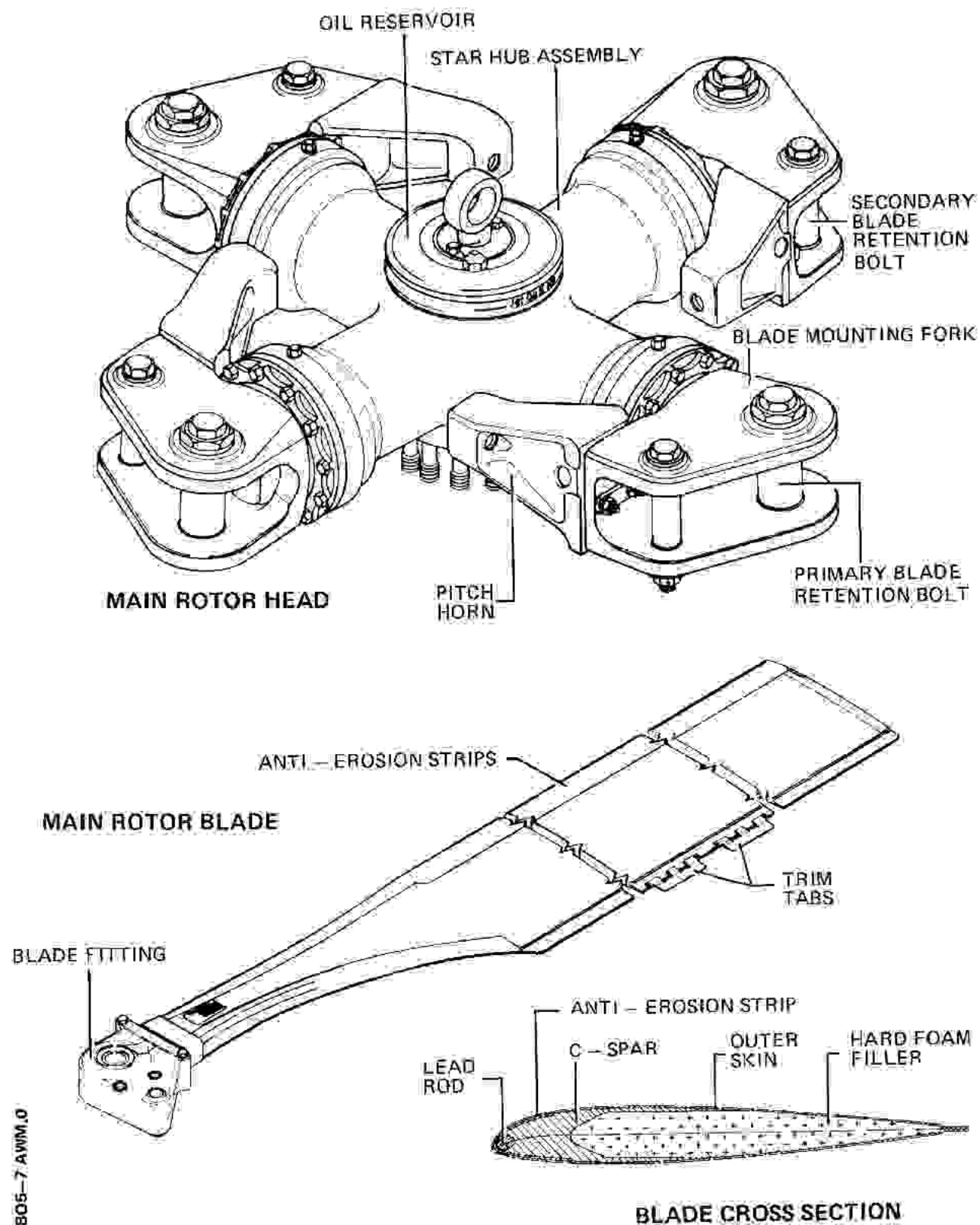


Fig. 7-11 Main Rotor

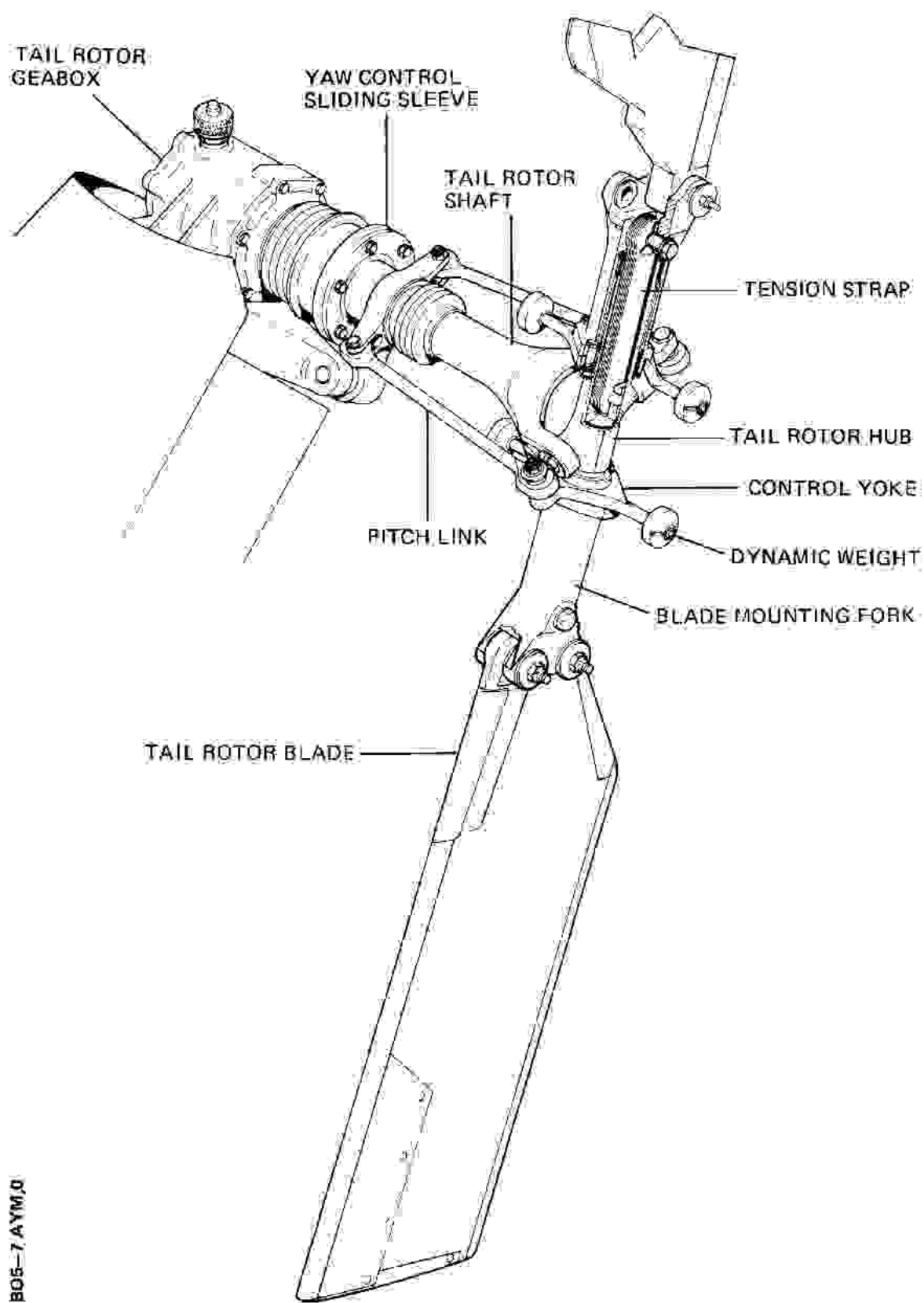


Fig. 7-12 Typical Tail Rotor Assembly

7.9 FLIGHT CONTROL SYSTEM (see Fig. 7-13)

The flight control system comprises a cyclic stick, a collective lever, and directional pedals to produce control inputs to the main rotor hydraulic servo actuators and tail rotor pitch change links.

Cyclic and collective servo actuator control inputs are combined by mixing levers and further transmitted to a swashplate. The swashplate changes the linear motion from the mixing levers to rotating motion and conveys pitch changes to the four main rotor blades.

Yaw control inputs are transmitted to the tail rotor sliding sleeve. The pitch of each blade changes when directional control inputs cause the nonrotating portion of the sliding sleeve to act upon the rotating portion. Pitch links between the sliding sleeve rotating portion and the pitch horn of each blade cause collective pitch angle changes of the tail rotor blades.

7.9.1 Cyclic Stick (see Fig. 7-13 and 7-14)

The cyclic stick is used to induce helicopter movements about the pitch and roll axes. The basic cyclic stick grip has switches for RPM-warning audio cutoff, cyclic control force trimming, and radio and internal communications microphone keying.

NOTE Instead of the new cyclic stick centering device (without locking possibility) in some H/C's a cyclic stick locking device is installed (observe ASB MBB-BO105-40-106). This locking device has a preset breaking point. It provides the possibility to override the locking device in case of emergency by a strong jerky force-aft movement of the cyclic stick (e.g. unintended locked cyclic stick in takeoff situation).

7.9.2 Collective Lever (see Fig. 7-13 and 7-14)

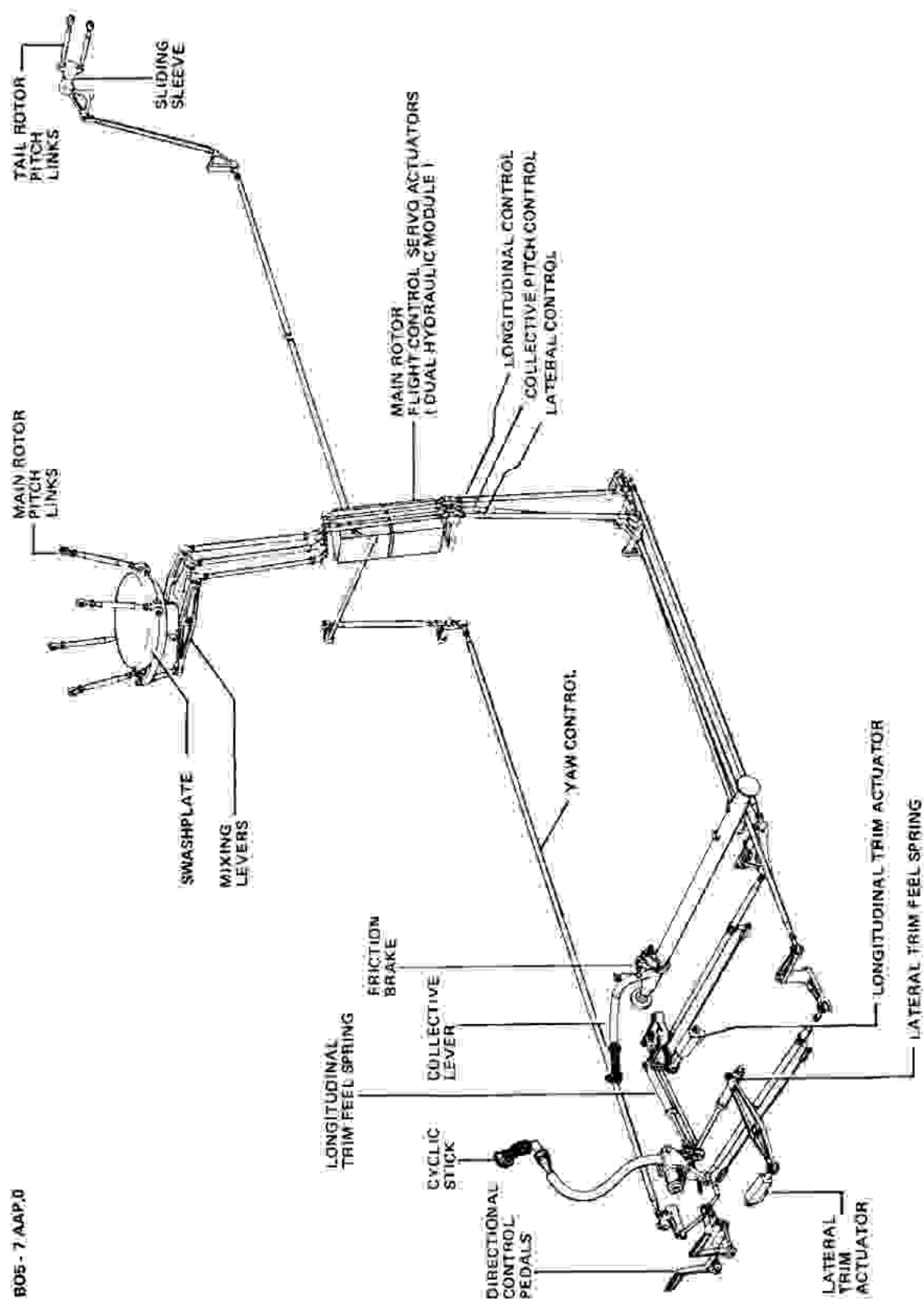
The collective lever provides a means of adjusting pitch angle of the main rotor blades and fuel flow metering requirements of the power turbine. A switch panel at the end of the lever has several switches installed, however, only the 4-way ENGINE TRIM switch and, on S-751 and subsequent, the LANDING LIGHT switches are active on the standard model. A friction adjustment at the base of the lever is used to prevent the collective pitch setting from moving when the lever is released.

7.9.3 Directional Control Pedals (see Fig. 7-13)

The directional control pedals are used to induce helicopter movement about the yaw axis. The pedals are adjustable fore-aft by use of a knob located aft of and between the pedals.

7.9.4 Cyclic Trim System (see Fig. 7-13 and 7-14)

The position of the cyclic stick may be electrically trimmed using the cyclic trim system. A lateral and a longitudinal trim feel spring assembly, each with an electrical actuator, are incorporated to produce artificial feel and cyclic control system trim. The spring assemblies will hold the cyclic stick in a neutral-feel trimmed position. Movement of the cyclic stick will cause the spring assemblies to compress and provide feel to the controls. When control pressure is released the cyclic will return to the neutral-feel position.



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Fig. 7-13 Flight Control System

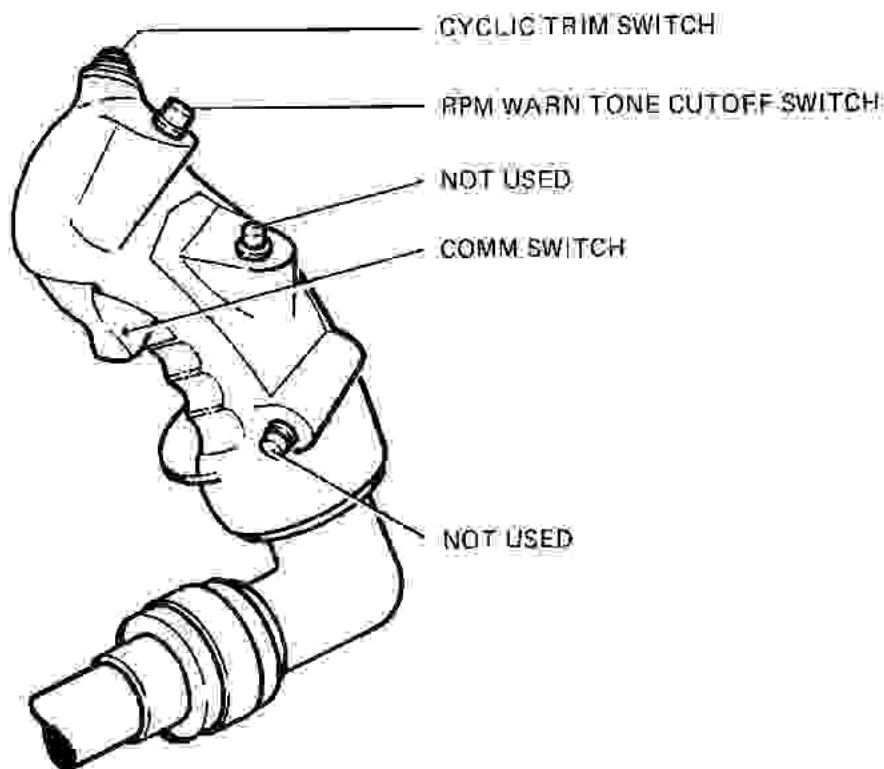
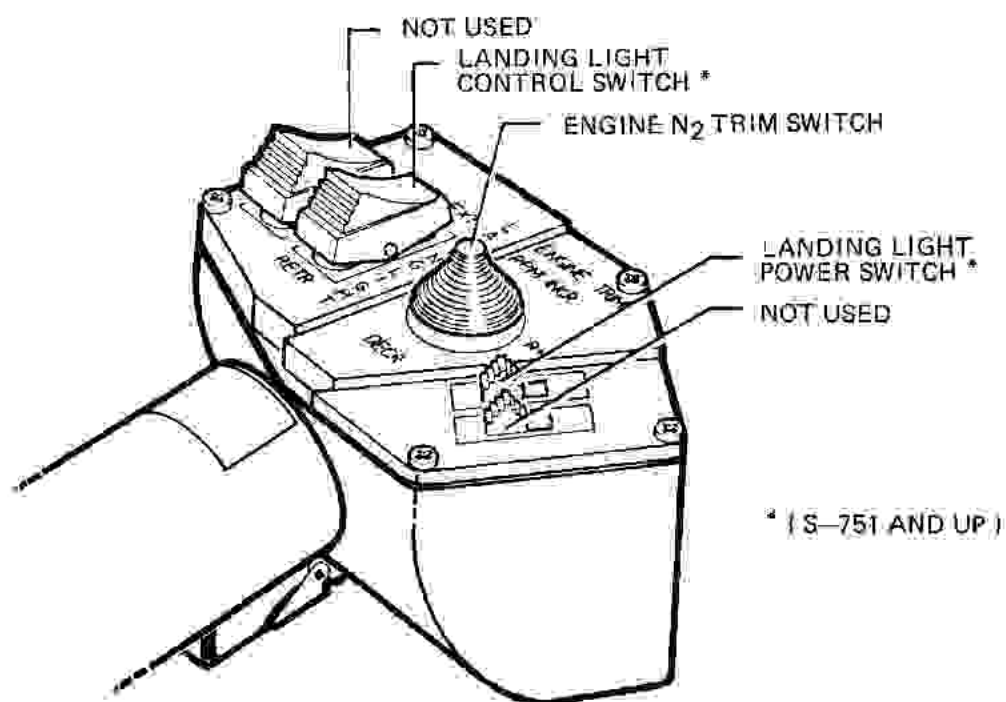
Manual retrimming of the neutral-feel position is accomplished by use of the 4-way TRIM switch on top of the cyclic stick grip. Pressing the switch in the desired direction will activate the appropriate trim actuator motor to reposition its respective spring assembly to establish a new neutral-feel position.

The system is powered direct from the Main Bus via the TRIM ACTUATOR circuit breaker on the center console switch panel.

7.9.5 **Flight Control Servos** (see Fig. 7-13 and 7-16)

The flight control servos are utilized to amplify the flight control forces applied to the main rotor system and thereby ease pilot workload.

Each of the three main rotor control axes has a tandem hydraulic servo actuator comprised of a primary servo and a standby servo. These servos are independent of each other to provide redundancy in the flight control system. Should one hydraulic system fail the remaining system can drive the main rotor flight control servos.



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Fig. 7-14 Cyclic Stick Grip and Collective Lever Controls

7.10 HYDRAULIC SYSTEM (see Fig. 7-15 and 7-16)

The hydraulic system, designed to reduce the main-rotor control forces, is constructed of modular components and mounted on the left side of the transmission compartment. It consists of two independent but mainly identical systems, of which System 1 is the primary operating system while System 2 is the stand-by reserve system. If necessary, operational control crosses over from the primary system to the stand-by system automatically and without delay. Both systems offer 100 % specific power support.

Pressure supply is provided by two hydraulic pumps which are permanently driven by the main transmission and which normally deliver full system pressure. The control of pressurization for the three actuators in each system is provided by two selector valves (one each system) which are linked by an interconnecting plate to ensure that only one system is active. The selector valves are mechanically coupled with and controlled by a hydraulic shuttle valve that works on the basis of pressure differentiation. Should the pressure of the primary system (System 1) fall below a certain level, the shuttle valve will position the selector valves to cause the stand-by system (System 2) to take over control (cross-over) and System 1 to become inactive.

The pressure of each hydraulic system is monitored by a pressure switch that, if system pressure falls below approx 70 %, will activate the respective **HYDR 1** or **HYDR 2** warning light.

EFFECTIVITY *On hydraulic systems with Positive Cross-over Indication:*

The **HYDR 1** warning light will also come on (together with the **HY BLOCK** warning light) whenever System 2 is activated.

EFFECTIVITY ALL

A jammed servo valve in System 1 will cause an electric signal to be released by micro switches which, by means of a relay, trigger a solenoid valve. This solenoid valve causes an artificial loss of pressure by opening a reverse flow in System 1 so that System 2 is activated by the shuttle valve and System 1 is then deactivated (cross-over). The **HY BLOCK** warning light comes on whenever the relay controlling the solenoid valve is activated.

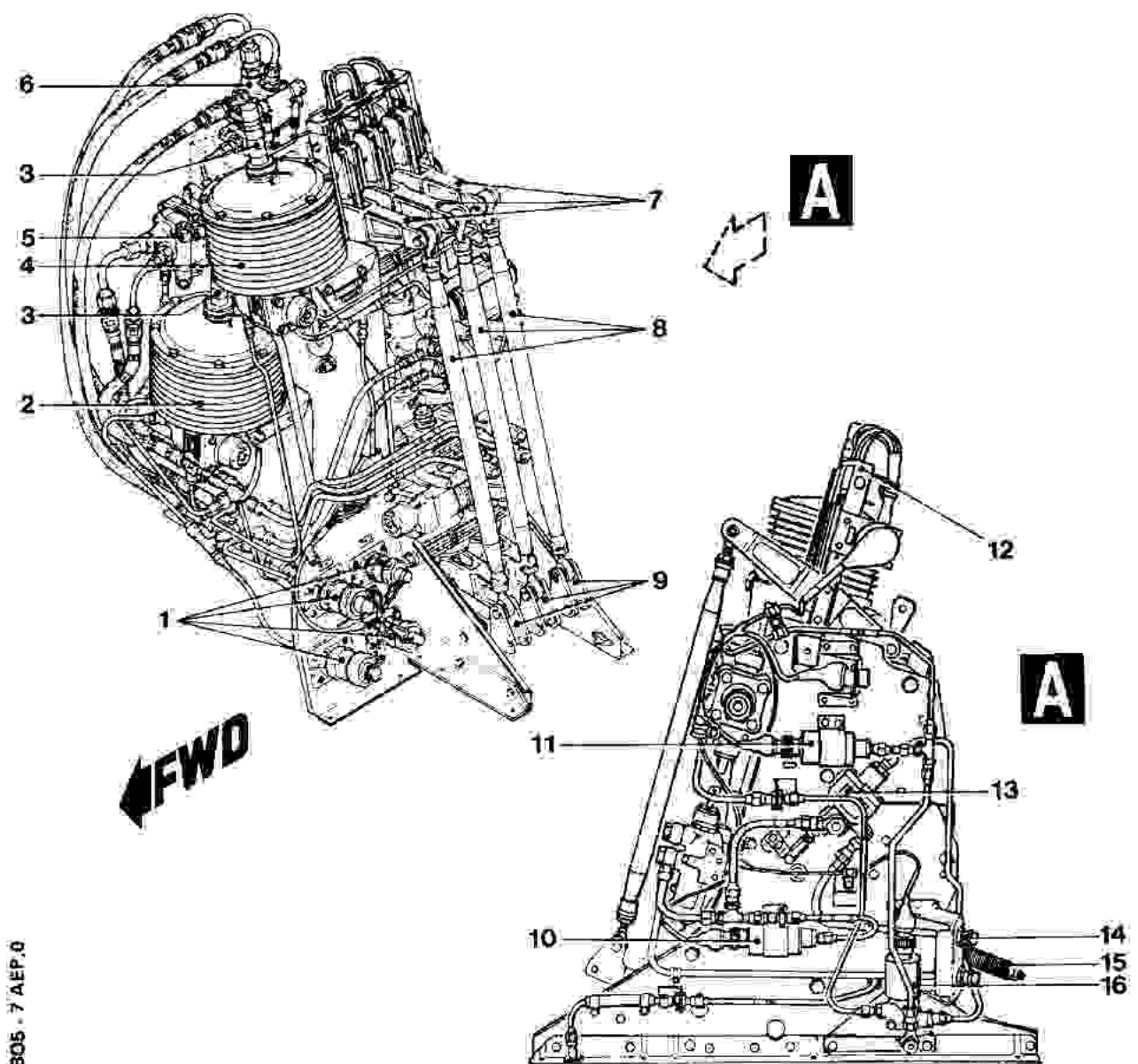
The micro switches, noted above, are designed to recognize a jammed servo valve and are only integrated in System 1. In System 2, however, a connection provided by springs will by-pass a jammed servo valve noticed by increased control forces about the affected axis.

NOTE Should a servo valve jam in System 2, an automatic cross-over to System 1 is impossible. Therefore, normal operation in System 2 is not allowed as the then lack of redundancy would cause an inflight system failure to be a safety risk.

A cross-over functional check can be performed using a guarded three-position **HYDR TEST** switch located on the center console switch panel. Placing the switch from the

spring-loaded center (off) position to the TEST position activates the relay and, as a result, the solenoid valve to cause control to cross-over to System 2. Placing the switch to the RESET position deactivates the solenoid valve causing control to revert back to System 1.

All system control and monitoring functions are powered direct from the Main Bus via the HYDR. CONTROLS circuit breaker on the center console switch panel.



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- | | |
|--------------------------------|---------------------------------|
| 1 SERVICE COUPLINGS | 9. INPUT LEVERS - SYSTEM 2 |
| 2 RESERVOIR - SYSTEM 2 | 10 PRESSURE SWITCH - SYSTEM 2 |
| 3 FLUID LEVEL INDICATORS | 11 PRESSURE SWITCH - SYSTEM 1 |
| 4 RESERVOIR - SYSTEM 1 | 12 MICRO SWITCHES |
| 5 HYDRAULIC PUMP - SYSTEM 2 | 13 SHUTTLE VALVE |
| 6 HYDRAULIC PUMP - SYSTEM 1 | 14 OPTIONAL EQUIPMENT COUPLINGS |
| 7 INPUT LEVERS - SYSTEM 1 | 15 DROOP COMPENSATION ASSEMBLY |
| 8 INTERCONNECTING CONTROL RODS | 16 SOLENOID VALVE |

Fig. 7-15 Dual Hydraulic Module

MANUFACTURER'S DATA

Rev. 0

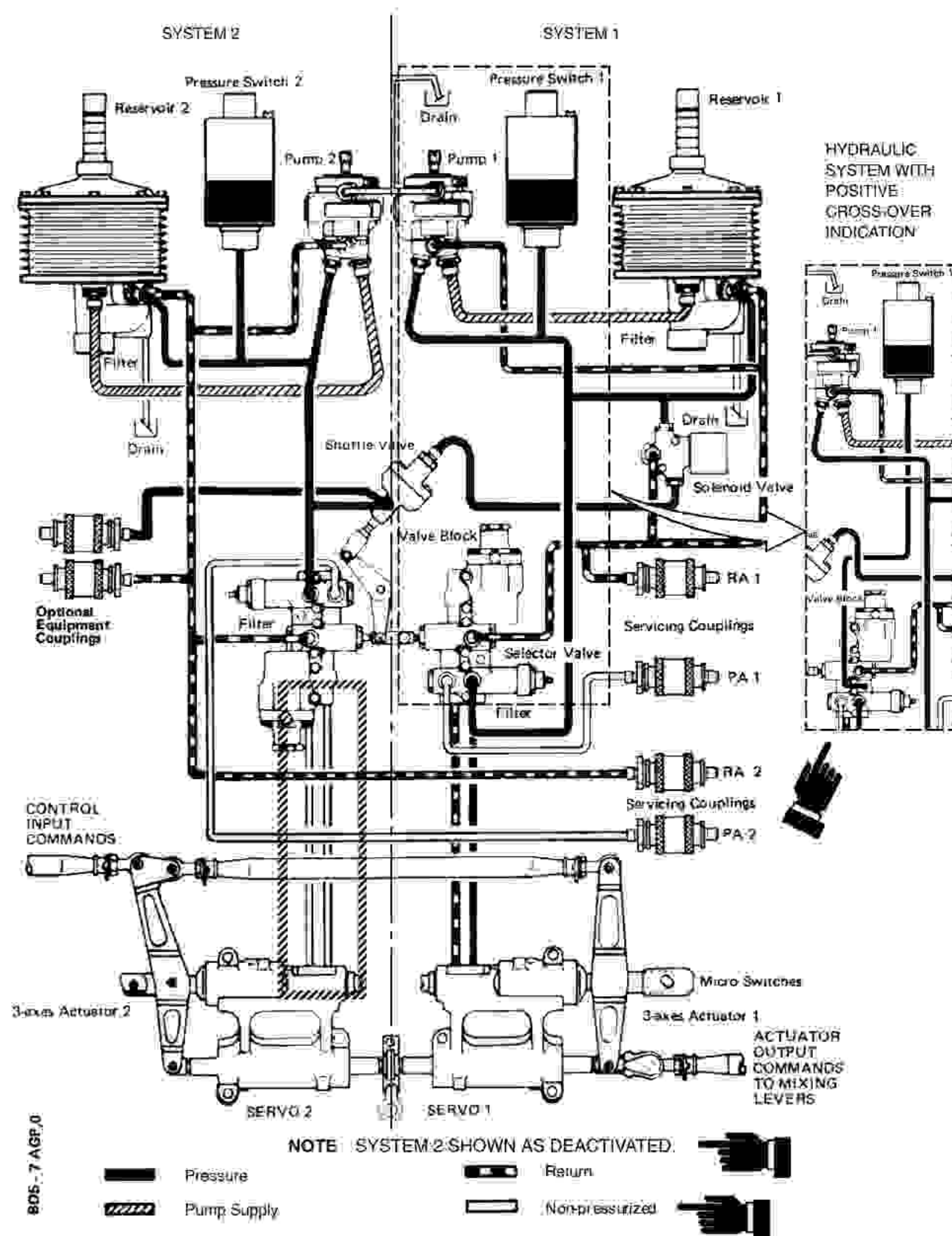


Fig. 7-16 Hydraulic System Schematic

7.11 ELECTRICAL SYSTEM

The electrical system provides 28V DC electrical power to the helicopter Starter, Main and Auxiliary Buses. An onboard 24V DC nickelcadmium battery supplies engine starting power and serves as an inflight emergency power source should both generators fail.

7.11.1 Power Supply (see Fig. 7-17)

The power supply comprises two engine-driven, 150-ampere DC starter/generators. The two generators, each with a voltage regulator and a differential reverse current relay, operate in parallel to supply 28V DC single-pole negative ground electrical power. Each voltage regulator controls the output voltage of its generator and equalizes the load between the two generators. The differential reverse current relay monitors generator output and will disconnect the generator from the system if its output voltage is less than the battery voltage and will cause the respective **GEN** warning light to come on.

7.11.2 Distribution (see Fig. 7-17)

Electrical power is distributed through use of three primary buses:

- Starter Bus primary power distributor that supplies battery and/or external power for engine starting, generator power for equipment requiring high current loads and all sources of power for further distribution to other buses.
- Main Bus supplies power to all equipment and systems essential for normal helicopter operation. This bus further supplies two Radio Nav Buses that power the COMM/NAV equipment.
- Auxiliary Bus supplies power to all non-essential consumers only when both generators are operational or when an external power source is connected.

7.11.3 Controls, Indicators and Annunciators (see Fig. 7-17)

STARTER/GENERATOR, ENG 1/ENG 2 switches when placed in the **GEN ON** position allow control voltages from the respective generator to flow to the reverse current relays. When the control voltage is greater than the battery voltage the relay will switch the generator on-line and the appropriate **GEN** warning light will go off. During external power assisted starts, the generators are not brought on-line regardless of STARTER/GENERATOR switch position as long as the external power source is providing power to the system.

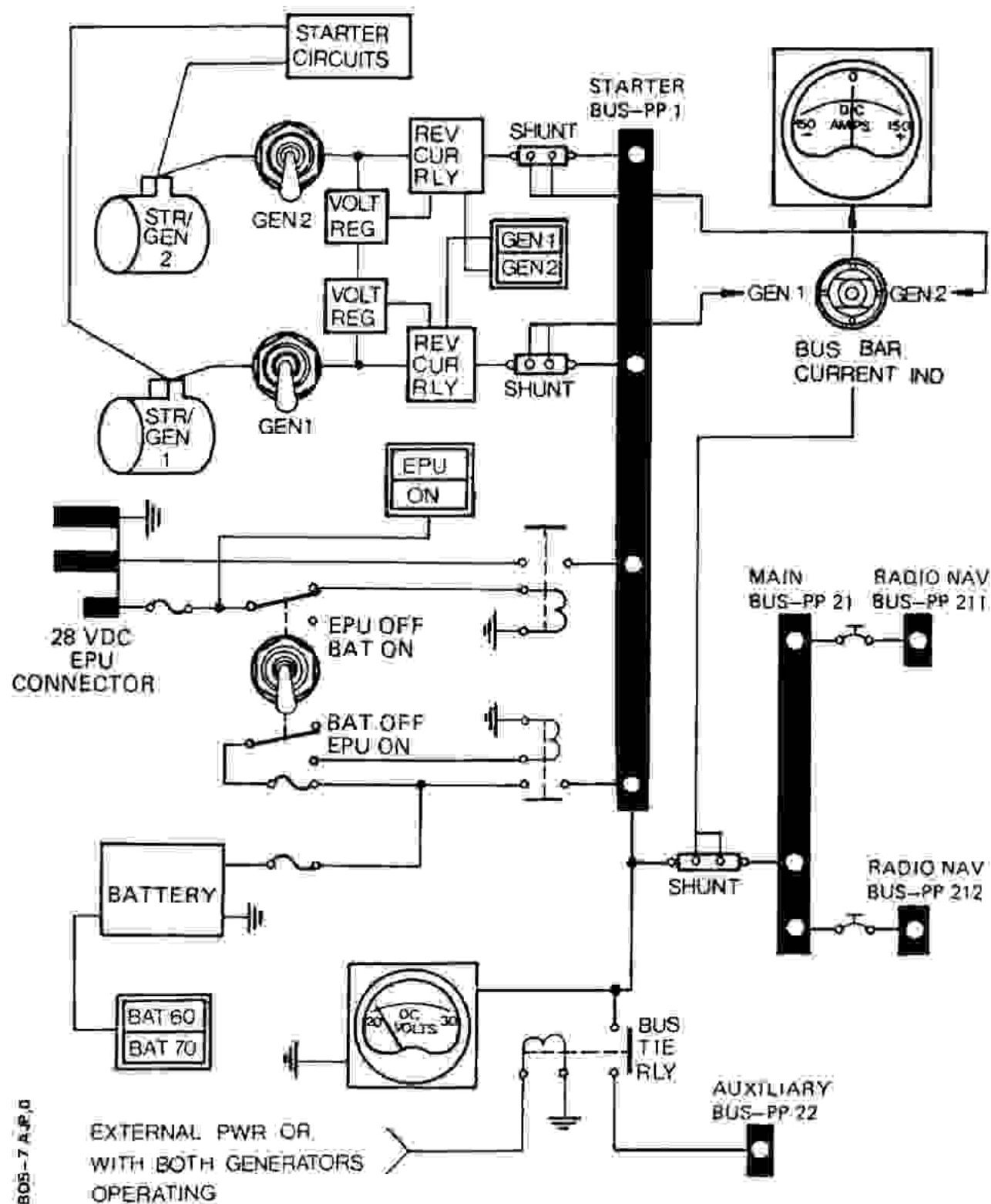
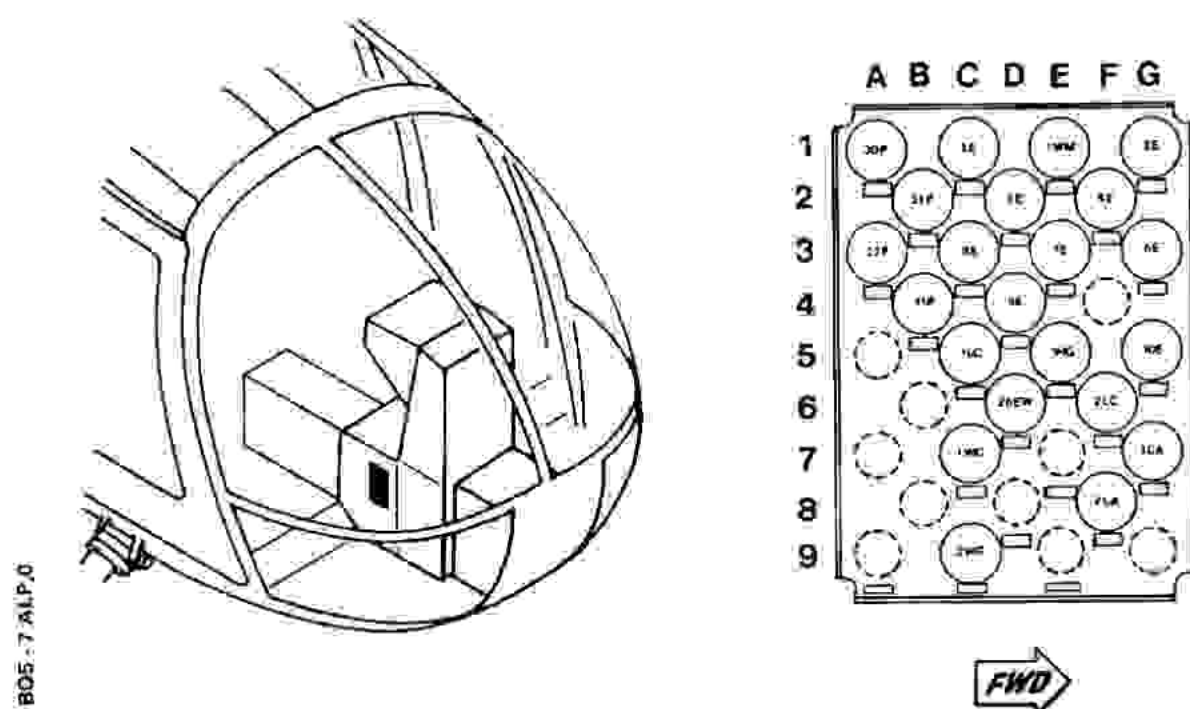


Fig. 7-17 Electrical System Schematic



FUSE IDENT	PANEL LOCATION	RATING	CIRCUIT(S) PROTECTED
2E	1G	1A	LOW FUEL warning
3E	1C	1A	Fuel quantity indicator
4E	2F	1A	RPM warning system
5E	2D	1A	Triple oil pressure indicator
6E	3G	1A	Triple oil temperature indicator
7E	3E	1A	Fuel pressure indicator
8E	3C	4A	Warn It test, T PLUG and OIL COOL warnings (optional)
9E	4D	1A	Fuel FILT 1/FILT 2 warnings
10E	5G	1A	Engine MAG PLUG 1/MAG PLUG 2 warnings
26EW	6D	1A	Transmission T OIL warning
1HG	5E	3A	Pitot tube heating
1LA	7G	10A	Anticollision lights
2LA	8F	4A	Position lights
1LC	5C	3A	Instrument lights
2LC	6F	4A	Interior lights and utility light
11P	4B	4A	Battery control relay
30P	1A	4A	EPU control and EPU ON caution light
31P	2B	4A	Ammeter main bus shunt - negative pole
32P	3A	4A	Ammeter main bus shunt - positive pole
1WG	7C	1A	Fire warning - Engine 1
2WG	9C	1A	Fire warning - Engine 2
1WM	1E	2A	Mast moment indicator system

Fig. 7-18 Typical Fuse Panel Layout

Battery switch, when placed in the BAT ON/EPU OFF position, connects the onboard battery to the Starter and Main Buses. Battery charging is then supplied by the generators when they are brought on-line. When the battery switch is placed in the BAT OFF/EPU ON position a 28V DC external power unit (EPU) may then supply power to the Starter Bus which in turn powers both the Main and Auxiliary Buses. An EPU ON caution light comes on when external power is connected and switched on. The EPU receptacle in the nose compartment allows for a starting current of 700 A.

BAT 60 and BAT 70 warning lights are provided to indicate an overheating condition of the battery. These lights operate independently being activated by separate battery-mounted temperature sensors.

Voltmeter (DC VOLTS) indicator indicates the voltage output supplied by the Main Bus during generator, battery or external power operation.

Ammeter (DC AMPS) indicator with the CURRENT IND switch placed in the BUS BAR position indicates current consumption (– side) measured between the Starter Bus and Main Bus, and with the CURRENT IND switch in either 1 GEN or 2 GEN position will indicate the current supplied (+ side) by the respective generator.

7.11.4 Circuit Protection

All electrical power circuits are protected by either a circuit breaker or fuse. Most circuit breakers are located on the center console switch panel. Most low amperage fuses are found on a fuse panel (see Fig. 7-18) located behind a cover on the instrument panel lower RH structure. Fuses and circuit breakers used to protect high amperage consumers and the generators are found in a main relay box mounted to the fuselage RH structure in the cargo compartment.

7.12 LIGHTING EQUIPMENT

The lighting equipment comprises exterior and interior lighting systems and the annunciator lights. The exterior lighting system includes the position lights, the landing light and the anticollision lights. The interior lighting system includes the instrument lights, a utility light, and the internal compartment lights.

7.12.1 Position Lights

The position lights are located on the horizontal stabilizer end-plates (RH – green, LH – red) and on the top aft end of the vertical fin (white). Power for operation of the lights is provided by the Main Bus via a fuse. Operation of the lights is controlled by the POSITION LIGHT switch on the overhead console.

7.12.2 Anticollision Lights

Two high-intensity flashing red anticollision lights are located, one on top of the tail rotor gearbox, the other on the underside of the fuselage. Power for operation of the lights is provided by the Main Bus via a fuse. Operation of the lights is controlled by the ANTI-COLL LIGHT switch on the overhead console.

7.12.3 Instrument Lights

Power to the instrument lights is controlled by the INSTR LIGHT switch on the center console switch panel. With this switch on power is supplied from the Main Bus via a fuse to a dim control unit. The light intensity is then controlled by the INSTR LIGHT rheostat switch located on the instrument panel.

A whip light at the left of the center console switch panel is also powered when the INSTR LIGHT switch is on. The whip light, however, has its own on-off switch and the light intensity is not effected by the rheostat switch.

7.12.4 Utility Light

A utility light is provided as a detachable light with coiled extension cord located at the left of the overhead console. This light is operated by an OFF-BRT rheostat switch integral with the light unit and is independently powered by the Main Bus via a fuse.

7.12.5 Interior Lights

A cabin light and two cargo compartment lights are powered from the Main Bus via the same fuse provided for the utility light. Operation of the three ceiling lights is controlled by the INTERNAL LIGHT switch on the overhead console.

7.12.6 Annunciator Lights (see Fig. 7-19)

When the TEST WARN. LIGHTS pushbutton on the instrument panel is pressed, all warning/caution lights (except the two F [Fire] and the LIMIT [MM Ind] lights) will come on. Each warning/caution light segment has two lamps so that failure of a single lamp will produce only a dimming effect. If a segment fails to light, it indicates either the failure of both lamps or a defective circuit.

7.12.7 Rpm Warning System

Visual and in most cases audio signals are provided as a warning whenever:

- N_{R0} is above or below the normal operating range.
- an engine N_1 differential equal to or greater than 12 % occurs (after either power lever is in the FLIGHT position).

The engine N_1 differential warning may be cancelled by pressing the RPM WARN pushbutton on the cyclic stick grip.

WORD SEGMENT	COLOR	ILLUMINATION PARAMETER OR FAULT
ANTI-ICING 1	Amber	No. 1 engine anti-icing actuator in open position.
ANTI-ICING 2	Amber	No. 2 engine anti-icing actuator in open position.
BAT 60	Red	Battery temperature exceeds 60 °C.
BAT 70	Red	Battery temperature exceeds 70 °C.
EPU ON	Amber	External power unit connected and switched on.
F (LH)	Red	Fire detected in the No. 1 engine compartment.
F (RH)	Red	Fire detected in the No. 2 engine compartment.
FILT 1	Red	No. 1 fuel filter is clogged and bypass has begun.
FILT 2	Red	No. 2 fuel filter is clogged and bypass has begun.
GEN 1	Red	No. 1 generator is not on-line.
GEN 2	Red	No. 2 generator is not on-line.
HY BLOCK	Red	Jammed servo valve in hydraulic system no. 1 and cross-over to system no. 2 has occurred.
HYD 1	Red	No. 1 hydraulic system pressure is low and cross-over to system no. 2 has occurred.
HYD 2	Red	No. 2 hydraulic system pressure is low.
LIMIT	Red	Max mast moment limit exceeded.
LOW FUEL	Red	Fuel supply tank usable quantity is below 60 kg.
MAG PLUG 1	Red	No. 1 engine oil contains metal fragments.
MAG PLUG 2	Red	No. 2 engine oil contains metal fragments.
OIL COOL	Red	(After SB 60-37) Oil cooler fan failure.
RPM	Red	N_{R0} below 95% or above 103%; N_1 differential greater than 12%.
T OIL	Red	Transmission oil pressure below 0.5 kp/cm ² ; or oil temperature above 105 °C.

Fig. 7-19 Alphabetical Listing of Warning/Caution Lights

The visual signal will always be indicated by a **flashing RPM** warning light. The audio signals heard through the pilot headset earphones vary depending on the fault condition as follows:

FAULT CONDITION	AUDIO SIGNAL
N_{R0} Low - 75 to 95 %	Beeping tone
N_{R0} High - 103 %	None
N_{R0} High - 108 %	Steady, high-pitched tone
N_1 Differential $\geq$ 12 %	Beeping tone

NOTE The N_{R0} High - 108 % RPM warning applies only for those helicopters equipped with a two-stage warning unit.

7.13 FLIGHT INSTRUMENTS

The instruments discussed under this subject are, for the most part, those that directly measure flight performance.

7.13.1 Pitot-static System

An electrically heated pitot tube is installed at the fuselage lower RH nose area. Two associated static ports are installed flush with the fuselage; one on the right and one on the left. The pitot tube supplies ram air to the airspeed indicator. The pitot heater is powered from the Main Bus via a fuse. The PITOT HEATING switch on the center console switch panel controls the heating element as indicated by the PITOT HEATING indicator on the instrument panel. The static ports provide static air pressure to the barometric altimeter, vertical speed indicator and airspeed indicator.

7.13.2 Barometric Altimeter

The barometric altimeter reacts to changes in barometric pressure delivered via the static ports. The altimeter scale is graduated in fifty-foot increments and numbered at hundred-foot intervals. The long needle indicates 100 foot intervals, the short needle indicates 1000 foot intervals and the rotating scale to the right of center indicates 10,000 foot intervals. The scale window at the lower center of the instrument face indicates barometric pressure setting in hectopascals (millibars) adjustable by use of the knob on the lower left corner of the instrument case.

7.13.3 Vertical Speed Indicator

The vertical speed indicator presents the rate of climb or descent from 0 to 6000 feet per minute. The instrument scale is graduated in 100 feet per minute increments from 0 up to 1000 feet per minute, and in 500 feet per minute increments from 1000 to 6000 feet per minute.

7.13.4 Airspeed Indicator

The airspeed indicator presents airspeed from 0 to 150 knots. The indicator dial is scaled in five-knot increments above 10 knots.

7.13.5 Clock

The mechanical clock on the instrument panel is wound and set with a knob at the lower center of the clock face. Below the knob is a lever used for starting, stopping and resetting (zeroing) an elapsed-time counter. A small lever to the right of the knob is used for stopping the sweep second-hand.

7.13.6 Standby Magnetic Compass

The standby magnetic compass is attached to the windshield center frame above the instrument panel. A compass correction card is affixed to the windshield frame above the compass.

7.13.7 Outside Air Temperature (OAT) Indicator

The OAT indicator is a self-contained unit mounted to the RH cabin roof window. A probe extends through the window to sense outside free air temperature. The dial has two scales to indicate temperature in degrees Celsius and degrees Fahrenheit.